RFS | Australian UAV Service

Standard Operating Procedures
Uncrewed Aerial Vehicles (UAVs)

Version	1.2
Approver	Oliver Heys
Review Date	17/2/2025

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2 Terminology

Please see below some common names for UAV and associated terminology.

AUAVS	Australian UAV Service – Surf Life Saving Services
BVLOS	Beyond Visual Line of Sight (Requires CRP or SBP Approval)
CASA	Civil Aviation Safety Authority
CRP	Chief Remote Pilot
DCRP	Deputy Chief Remote Pilot
EVLOS	Extended Visual Line of Sight (Requires CRP or SBP Approval)
RFS	NSW Rural Fire Service
LEO	Location Equipment Officer
ReOC	Remote Operator's Certificate
RePL	Remote Pilot License
RPAS	Remotely Piloted Aircraft Systems
SAR	Search and Rescue
SBP	Senior Base Pilot
SOPs	Standard Operating Procedures
UAS	Uncrewed Aircraft System
UAV	Uncrewed Aerial Vehicle
UAV OIP	UAV Operator Induction Program
VLOS	Visual Line of Sight

3 Revision Log

Date	Version	Affected Parts	Affected sections	Summary Of Revision	Authorised By
6/12/2024	V1.0	All	All	Initial Publication	Oliver Heys
7/1/2025	V1.1	UAV Job/Mission Planning	6.2 Priority/Scheduled Operational Job 6.8 Job approval timeframes 19.1.2 – Naming convention consistency	Added job submission procedure for Operational Tasking job submissions, compared to non-operational jobs. Updated timeframes for job approvals depending on job type (operational or training).	James Bassam
16/02/2025	V1.2	AVCRM Job Naming	6.1 Creation of Mission	Note: Use date of Operation for job name.	Isaac Hails

4 Policy and procedures

4.1 Applicability

These Standard Operating Procedures provide instructions for the operation and management of Uncrewed Aerial Vehicles (UAVs) and all persons involved in the operation of UAVs that are operating under RFS and AUAVS Standard Operating Procedures:

- UAV Pilot (RePL)

Award proficiency is a requirement for using any RFS RPAS. Both an online and practical proficiency must be completed annually by all Operators and Pilots between 1 July and 31 December, as outlined and scheduled annually by AUAVS. Any member who completes an OIP or RePL between 1 July and 31 December does not need to complete a proficiency until the following year. These SOPs are subject to change at any time and should be considered non-current once printed. The most up-to-date version is available via the RFS AVCRM portal.

These SOPs are aligned to the Standard Operating Conditions outlined in Civil Aviation Safety Regulations, Part 101, however, for ReOC Operations, the ReOC Operations Manual takes precedence in all circumstances. It must be noted that Part 101 regulations always take precedence over the Operations Manual and SOPS

4.2 RFS relation to Managed Service Provider

RFS has engaged AUAVS as a Managed Service Provider to support its operational needs. This includes the creation of Standard Operating Procedures (SOPs), provision of necessary assets, and delivery of training. AUAVS will update these SOPs annually, incorporating any minor adjustments and feedback provided by both RFS and AUAVS's own operational guidelines.

4.3 RFS Contacts

Name	Role/Organisation	Phone	Email
Emergencies	N/A	**000**	N/A
AUAVS Team	AUAVS Direct Contact	02 9457 1027	nswrfs@surflifesaving.com.au
Brett Hagan	Manager Operational Field Support RFS	0429 449 885	brett.hagan@rfs.nsw.gov.au
Cassie Lock	RPAS Capability	0417 358 704	cassie.lock@rfs.nsw.gov.au
James Bassam	AUAVS Operations Manager, Safety and Compliance	0419 544 563	jbassam@surflifesaving.com.au
Oliver Heys	Chief Remote Pilot AUAVS	0472 651 733	oheys@surflifesaving.com.au

Mark Atkins	Deputy Chief Remote Pilot AUAVS	0419 918 409	matkins@surflifesaving.com.au

5 Roles and Responsibilities

5.1 Responsibilities of Chief Remote Pilot

The Chief Remote Pilot is responsible for all operational matters and Remote Pilot (RP) training affecting the safety of operations.

The role and responsibilities of the Chief Remote Pilot are:

- Ensure that operations are conducted in compliance with the Civil Aviation Act and the Regulations;
- Maintain a record of qualifications held by each Remote Pilot (RP);
- Monitor and maintain operational standards, supervise and train RP(s) who work under both the authority of the Remote Operators Certificate (ReOC);
- Maintain a complete and up-to-date reference library of operational documents as required by Civil Aviation Safety Authority (CASA) for the class of operations conducted;
- Develop applications for approvals and permissions where required to facilitate company operations;
- Develop checklists and procedures relating to flight operations;
- Be the point of contact with CASA;
- Notify CASA prior of any change to the ReOC manual or its schedule.

5.2 Responsibilities of Deputy Chief Remote Pilot

The Deputy Chief Remote Pilot is selected by the Chief Remote Pilot and must be an existing Senior base Pilot to ensure they have a high level of competency with operating on the ReOC. This role allows operations to continue during periods of high workload for the Chief Remote Pilot and Deputy Chief Remote Pilot can also be selected to stand in for the Chief Remote Pilot when they are not contactable or on leave. The roles and responsibilities of the Deputy Chief Remote Pilot are:

- Ensure that operations are conducted in compliance with the Civil Aviation Act and the Regulations;
- Conduct internal flight approvals and applications to CASA where required to facilitate company operations;
- Ensure the Chief Remote Pilot is informed, via email, of all matters relating to the above;
- Help select, review and mentor Senior Base Pilots;
- Assist the Chief Remote Pilot with Senior Base Pilot Auditing Schedules;
- Review all records and systems to ensure compliance is maintained.
- Assist the Chief Remote Pilot with manual improvements.

NOTE: The CRP must contact CASA to inform them of the Deputy Chief Remote Pilot's contact details when they are to cover the CRP whilst on leave or not contactable for a specific amount of time.

5.3 Responsibilities of a Maintenance Controller

The Maintenance Controller (MC) is responsible for ensuring that RFS RPAS are properly maintained. Without limiting the duties and accountabilities of the MC, the MC must:

- control all RPAS maintenance, either scheduled or unscheduled
- keep records of all personnel permitted to perform maintenance on RPA, including details of their training and qualifications
- develop, enforce and monitor RPAS maintenance standards
- maintain a record of RPAS defects and unserviceability issues
- ensure that each item of equipment essential to the operation of RFSs RPA is serviceable before being released to service
- maintain a thorough technical knowledge of RPAS operating under the authority of the ReOC of the organisation and ensure that all maintenance activities are conducted IAW the procedures detailed in the aircraft overview section.
- investigate all significant defects in the RPAS
- monitor the failure rates of RPAS components and impose additional maintenance requirements as necessary to ensure the safety of operations.

5.4 Responsibilities of UAV Pilot

For the purposes of this SOP a 'UAV Pilot' includes a holder of a Remote Pilot Licence (RePL) and has attended any further UAV OIP components required by the RFS and AUAVS, and who is proficient. If you believe there to be an issue, feel free to check with the UAV Team.

The UAV Pilot in Command of the aircraft is responsible for:

- Ensure RePL and AROC is uploaded to AVCRM;
- Creation of pre and post flight logs via AVCRM;
- Check Aircraft Defect log prior to all operations;
- Conduct of flight in accordance with company procedures;
- Safe operation of the aircraft;
- Safe Battery Management per UAV Manual and SOPs;
- Acting in accordance with company procedures;
- Follow the Standard Operating Conditions as defined by CASA's legislation.
- Ensure thorough cleaning post flight is completed and batteries should be placed on charge and supervised to ensure their condition does not cause overcharging or overheating. Batteries are to be inspected prior to charging as per the Battery Management Procedure.

5.5 Responsibilities of Camera Operator, Spotter & Others

All camera operators, spotters and other persons involved in the operation of UAV controlled under the authority of the ReOC or Sub-2Kg SOP are required to comply with the procedures and direction set out in the UAV SOP.

6 UAV Job/Mission Planning

Select appropriate Base per location of the operation. Please amend by clicking 'Change Base' if this is incorrect.

All UAV Operations are required to be logged prior to any flight operations via the AVCRM online portal. Please see detail in this SOP on how to create an operation.

Website: https://nswrfs.rpa.avcrm.net/

The AVCRM portal allows greater visibility for the UAV team to manage UAV operations and assist you with UAV operations, whilst ensuring you meet your legal requirements. The AVCRM system manages all UAV mission/job planning, area approvals, UAV Operator and Pilot qualifications and competencies, maintenance, UAV time in service.

All UAV Operators and Pilots must create and start their own AVCRM job using a personal login. It must not be completed by another person that is not directly involved in the operations of the flights on that day.

6.1 Create a Job or Mission

Select relevant base to location.

Select the Job Manager Tab on the left-hand side of the screen and select create job. Give your job a name following this format, YYYYMMDD_LocationName_Client_PilotName. See an example below:

(Note: Use date of Operation for job name.)

20241001_TerryHills_NSWRFS _BrettHagan

- 20241001
  - Date, 1st of October 2024
  - Note: Use date of Operation for job name.
- TerryHills
  - Location, all one word
- NSWRFS
  - Client, all one word as spelt and specified on AVCRM Drop List.
- BrettHagan
  - Pilot Name, all one word.

6.2 Priority/Scheduled Operational Jobs

For any jobs that are operational responses to a live incident or formal tasking for a scheduled event i.e. hazard reduction activity;

Select the Job Type as '*OPERATIONAL*' and prefix the Job Name with the below:

- **PRIORITY_OPERATIONAL** _20241001_TerryHills_NSWRFS _BrettHagan
  - For live incident response operations
- **SCHEDULED_OPERATIONAL**_20241001_TerryHills_NSWRFS _BrettHagan
  - For planned operational activities in the coming days

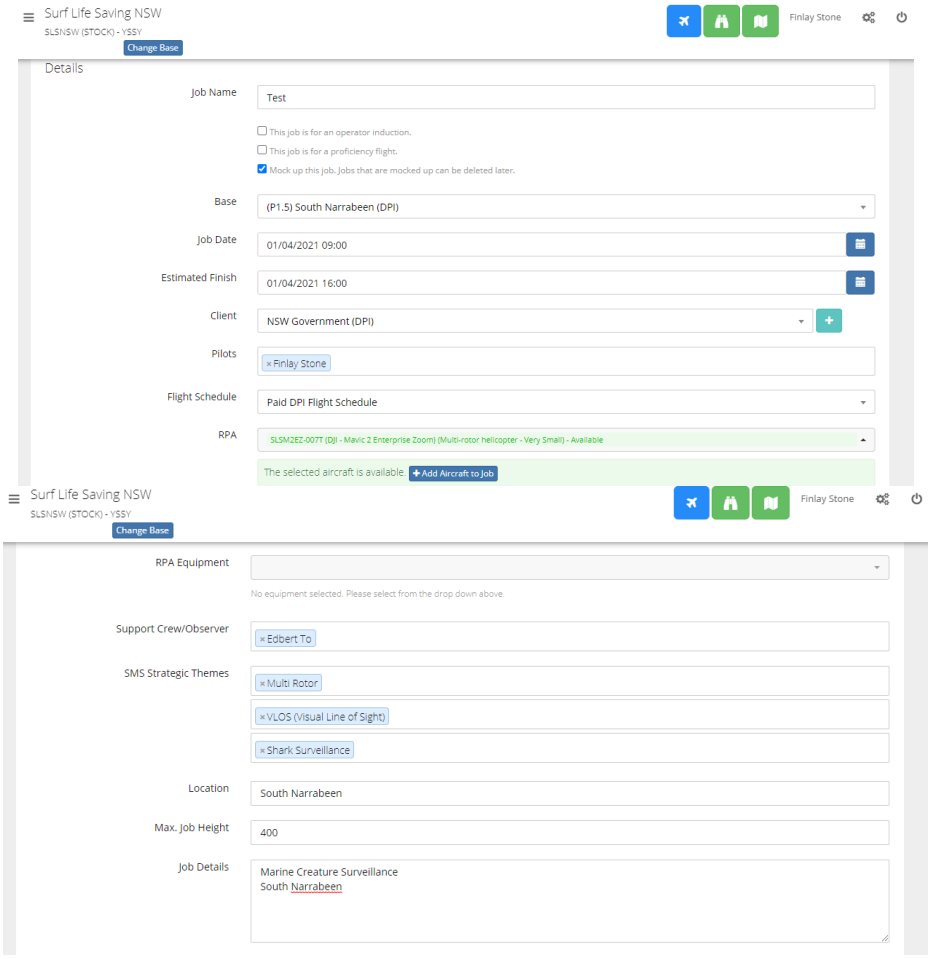


Important: Please follow up a Priority_Operational job with a phone call to the AUAVS support line (02 9457 1027) to advise of submitted priority job request. This ensures an efficient approval response time of less than 1 hour.

For non-operational jobs, i.e. familiarisation/demonstration/ training flights etc, select the Job Type as '*Training'*. No prefix is required in this instance. Non-operational job requests will be approved within 24 hours on a working day, so it is important to plan and submit these requests in advance.



Complete the remaining questions making sure all details are accurate including the pilot, as the AVCRM job will be logged against the nominated UAV Operator/Pilot.



6.3 Check Aircraft Defects

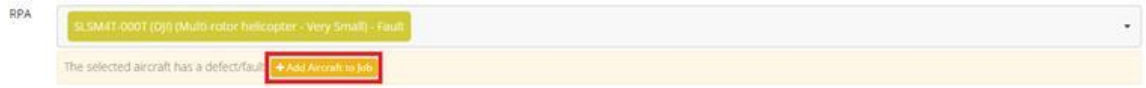
It is a requirement that all Pilots/Operators check for outstanding defects on aircraft prior to flying. Aircraft with defects are not to be flown.
You may see defects on aircraft which are related to batteries - in this case **the aircraft can still fly, but the identified battery or batteries must not be used**.

6.3.1 Please review the following procedure to easily check defects prior to every operation.

1. Select aircraft you plan on using for the job



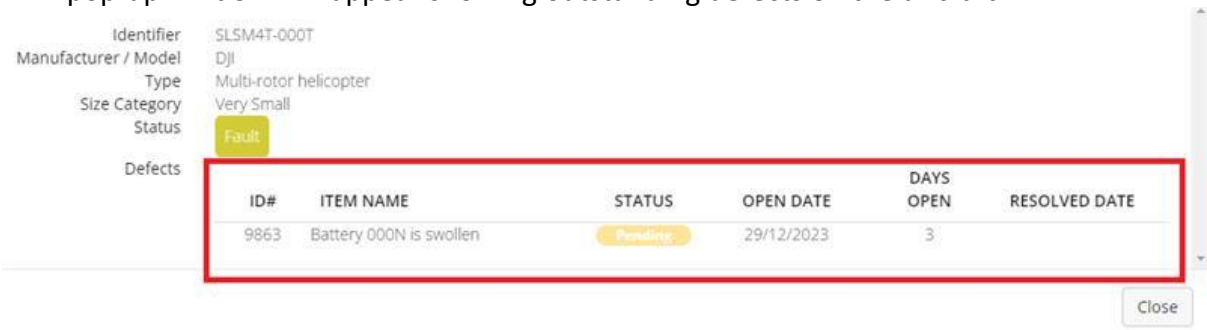
2. Click 'add to job'



3. Click on the newly added aircraft



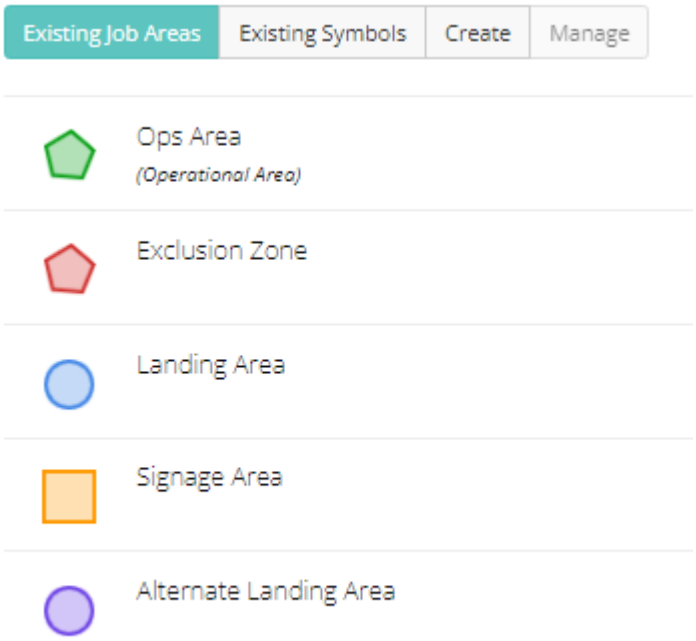
4. A pop-up window will appear showing outstanding defects on the aircraft



6.4 Job Area Creation

Highlight the area in which you wish to operate, include areas of UAV exclusion, the planned flight area, Take-off and Landing Area with alternative landing site and highlight where you are planning on placing your signage.



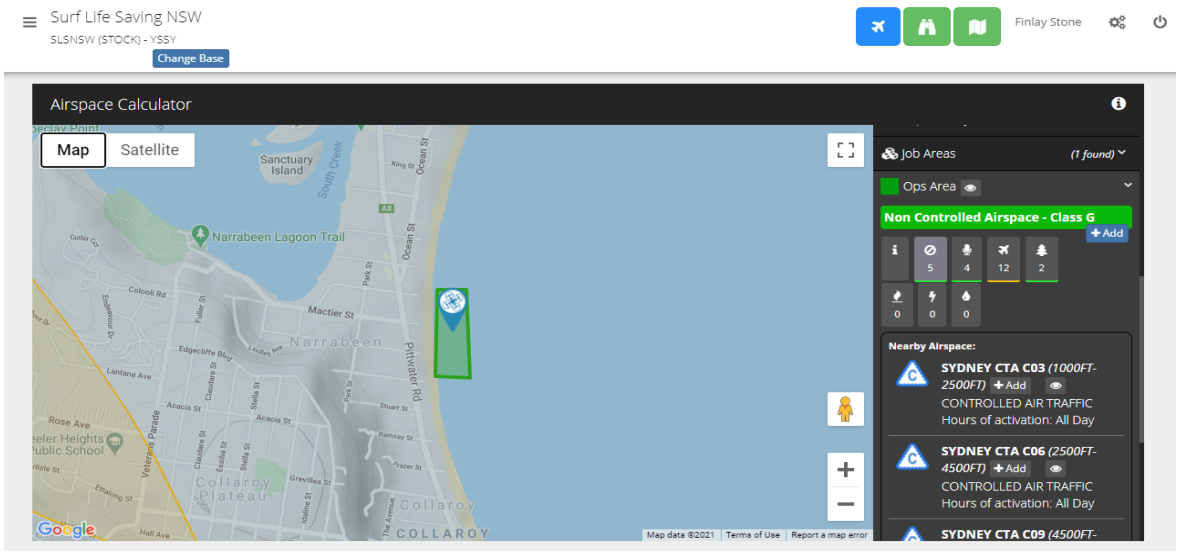


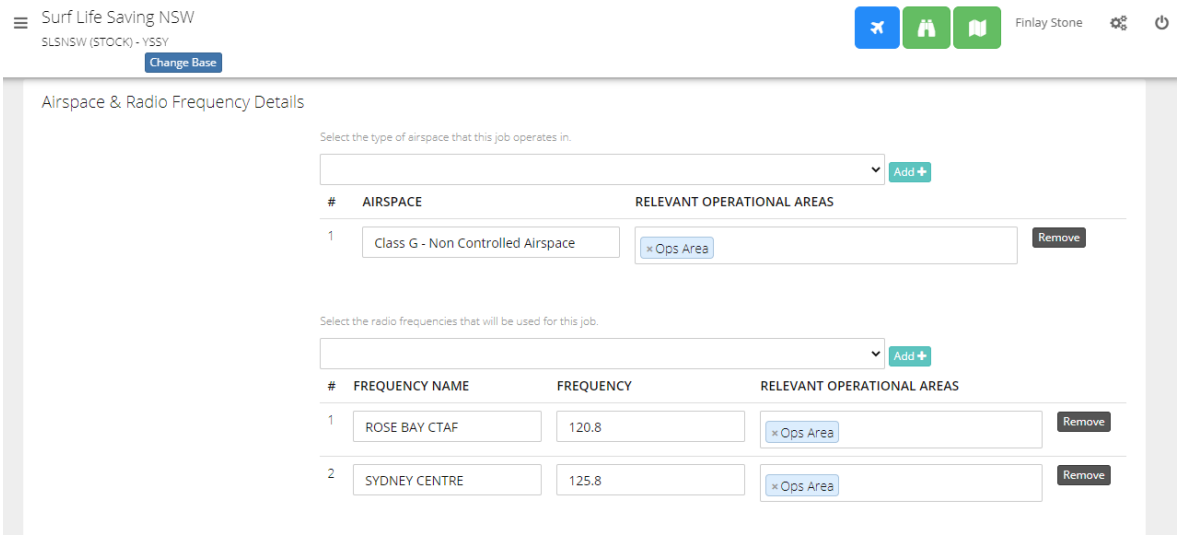
6.5 Check the Airspace Calculator

The Airspace Calculator will display your proposed job airspace on the left-hand side of the screen. The Airspace Calculator shows you the airspace you are operating in and around. Depending on your license type (Remote Pilot License or operating under CASA's Excluded Category), you can click on the coloured areas of airspace, restricted/danger areas, and other feature classes (National Parks, Fire and Emergency and other temporary restricted areas) and understand how you must operate.

By clicking on the Filter (top right of the calculator) you can expand your search radius and turn feature sets on and off.

By clicking the 'Add+' button on frequencies you can add the radio frequencies to your job. You can also find out necessary information of bearings and distances from these places.

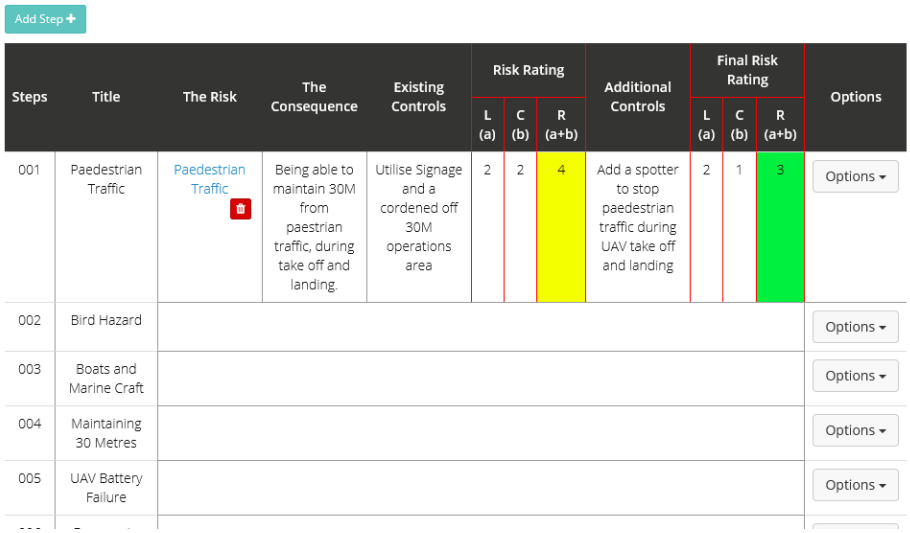




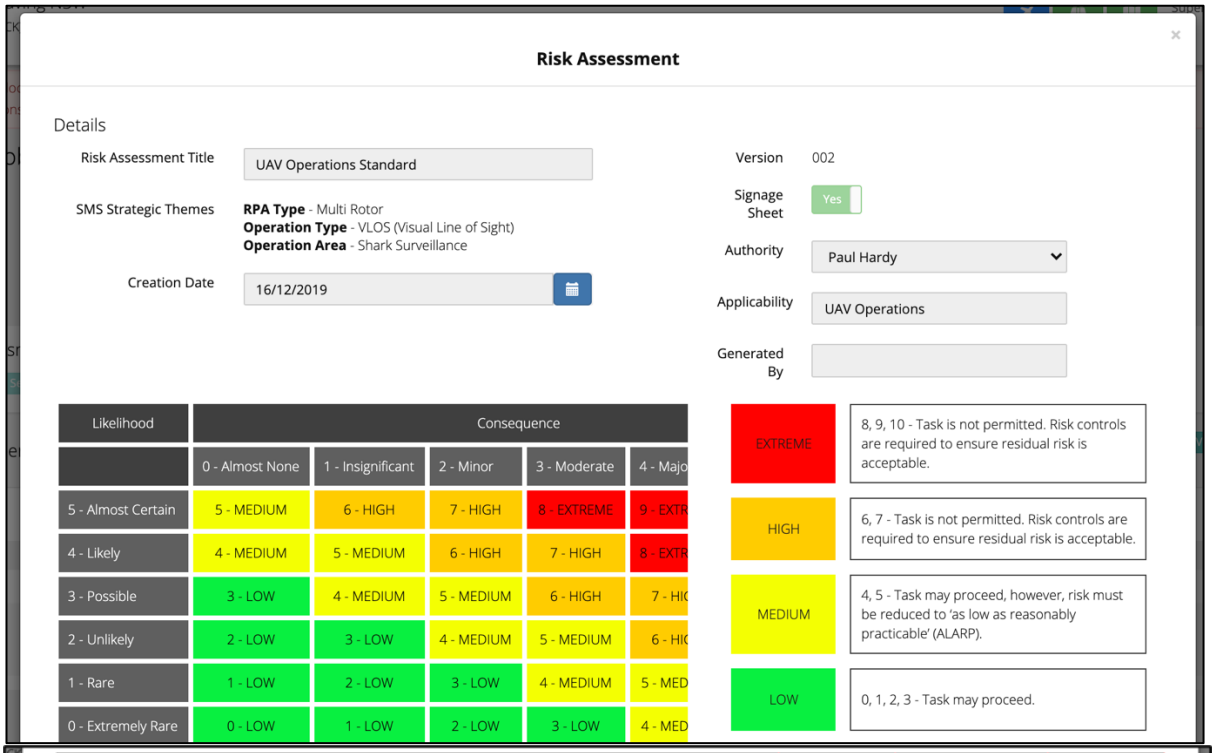
6.6 Risk Assessment

A risk assessment should be completed prior to any UAV operation. Your risk matrix will be pre-populated with common risks seen at the beach and how you can mitigate the risk. In the instance of RFS operational techniques detailed in the *Operational Techniques* section below, the UAV Operations Standard risk assessment is to be used. For operational types outside those detailed below, please contact the UAV team.

All you need to do is 'Select Existing' and add the *UAV Operations Standard* Risk Assessment to your job, using the most up-to-date version number.



You can then view the Assessment, to ensure its up to date, and relevant for the job.



Risk Assessment

Details

Risk Assessment Title	UAV Operations Standard	Version	002
SMS Strategic Themes	**RPA Type** - Multi Rotor **Operation Type** - VLOS (Visual Line of Sight) **Operation Area** - Shark Surveillance	Signage Sheet	Yes
Creation Date	16/12/2019	Authority	Paul Hardy
		Applicability	UAV Operations
		Generated By	

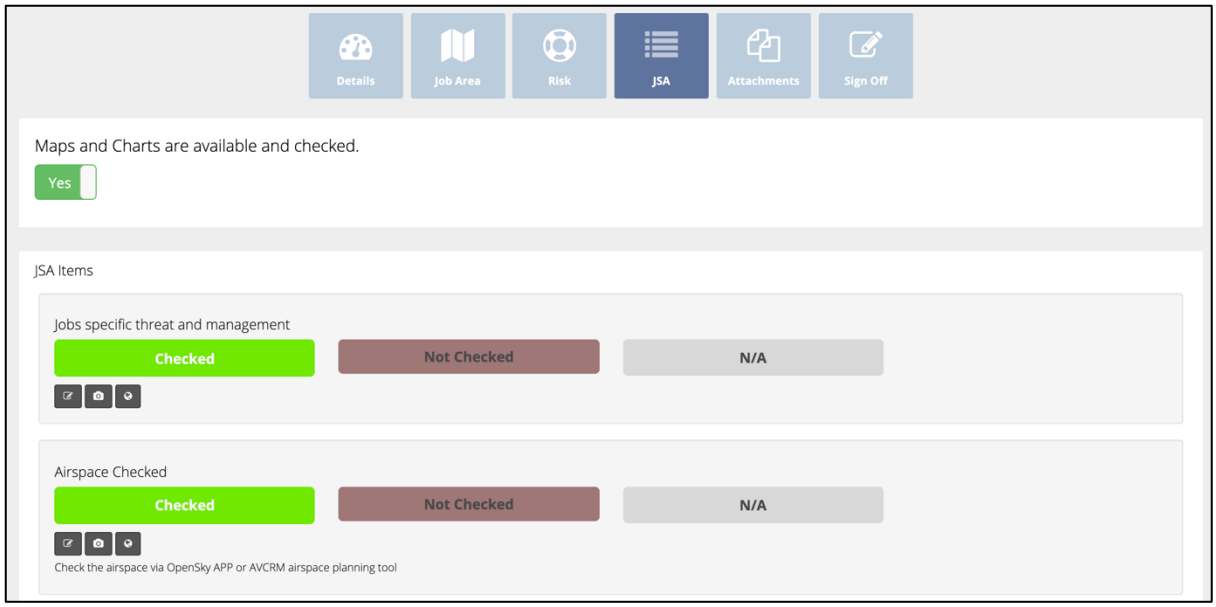
Likelihood	Consequence				
	0 - Almost None	1 - Insignificant	2 - Minor	3 - Moderate	4 - Majo
5 - Almost Certain	5 - MEDIUM	6 - HIGH	7 - HIGH	8 - EXTREME	9 - EXTR
4 - Likely	4 - MEDIUM	5 - MEDIUM	6 - HIGH	7 - HIGH	8 - EXTR
3 - Possible	3 - LOW	4 - MEDIUM	5 - MEDIUM	6 - HIGH	7 - HIG
2 - Unlikely	2 - LOW	3 - LOW	4 - MEDIUM	5 - MEDIUM	6 - HIG
1 - Rare	1 - LOW	2 - LOW	3 - LOW	4 - MEDIUM	5 - MED
0 - Extremely Rare	0 - LOW	1 - LOW	2 - LOW	3 - LOW	4 - MED

EXTREME	8, 9, 10 - Task is not permitted. Risk controls are required to ensure residual risk is acceptable.
HIGH	6, 7 - Task is not permitted. Risk controls are required to ensure residual risk is acceptable.
MEDIUM	4, 5 - Task may proceed, however, risk must be reduced to 'as low as reasonably practicable' (ALARP).
LOW	0, 1, 2, 3 - Task may proceed.

Steps	Title	The Risk	The Consequence	Existing Controls	Risk Rating			Additional Controls	Final Risk Rating		
					L (a)	C (b)	R (a+b)		L (a)	C (b)	R (a+b)
001	UAV SYSTEM FAILURE	RPA FAILURE	UAV failure causing the UAV to crash	As per exisinting Operational Manual: - Pre & Post flight inspections are carried out in line... read more	2	3	5	Maximise the amount of time the RPA spends flying over water and away from buildings or marine craft.	1	3	4
002	WILDLIFE (Birds of Prey)	WILDLIFE (BIRDS)	The RPA may be attacked and damaged/caused to crash.	Pre-flight inspection of the airspace and operational area.	3	2	5	Utilize a spotter, who can notify the RP via voice/local line of sight SLS radio; Return to home if... read more	1	2	3
003	ANIMALS (Dogs)	ANIMALS (DOGS)	Dog(s) may enter the operational area and: - Distract pilot; - Interfere with the RPA directly.	Existing controls from the Operations Manual: - The operational area marked with cones and signage... read more	4	1	5	Use of a spotter.	2	1	3
004	SAND & SALT EXPOSURE	SAND AND SALT EXPOSURE	Sand and salt (seaspray): - Corroding UAV parts leading to UAV failure and potential UAV crash; -... read more	Existing controls from the Operations Manual/SLSNSW Standard Operating Procedures: - Pre and Post... read more	3	4	7	RP to ensure any defects or concerns are logged in AVCRM, to be followed up on by the Authorise Maintenace Provider.	1	4	5
005	UAV BATTERY FAILURE	UAV BATTERY FAILURE	UAV battery fails, which causes a critical failure and the UAV to crash.	Existing controls from the Operations Manual/SLSNSW Standard Operating Procedures: - Pre and Post... read more	3	3	6	Make sure correct battery warnings are set in the UAV application if available. Make sure battery... read more	2	3	5

6.7 Job Safety Assessment (JSA)

Go through and complete the JSA as per layout, it is important to add any notes, photos or Geo Tags that are relevant to each part of the job.



Once the JSA is complete, you can save your job, ready for Starting.

6.8 Job Approval

Once you have saved your job it will be submitted to a Senior Base Pilot for approval.

Approval Timeframes:

- Formally tasked OPERATIONAL Job types (when followed up with a phone call) will have a <1 hour approval timeframe from submission 24/7.
- All other Non-Operational Job Types (listed as 'Training') will be approved within 24 hours on working days. Please plan these submissions in advance accordingly

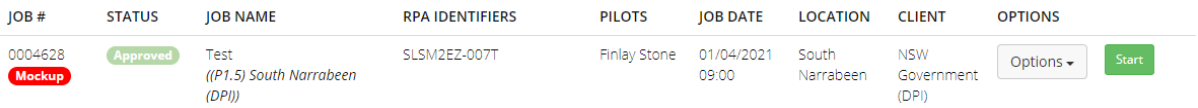
Please see Appendix 5: RPAS - Job Planning & Approvals Workflow for further information.

6.9 Pre-Flight Checklist

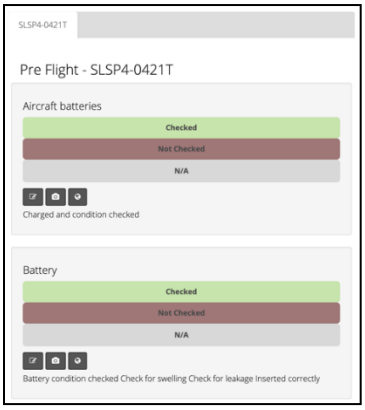
Complete and ensure that all items are marked as serviceable.

6.10 UAV Serviceability Prior to Operation AVCRM

Once ready, find your job under 'Job Manager'-'Manage Jobs', and click 'Start'. This will take you through the necessary information and allow you do complete the Pre Flight Checklist.



Pre-flight and post flight checks are mandated for all operations. The UAV Operator/Pilot must record the completion of these checks during the running stage of the job.



The Post Flight process is the same as Starting a Job, however, flight details must be added in, by syncing and linking flights (refer to pg. 26).

7 Defects

All defects found in the UAV must be recorded during the finish job stage in the AVCRM application. Once the job is completed the application will automatically enter the defect on that machines Electronic Defect/ Maintenance Log.

The UAV Operator/Pilot must ensure that all defects or outstanding maintenance actions detailed in the Defect/Maintenance Log have been addressed prior to operation of the UAV.

7.1 Knowing Availability Types for RPA Selection

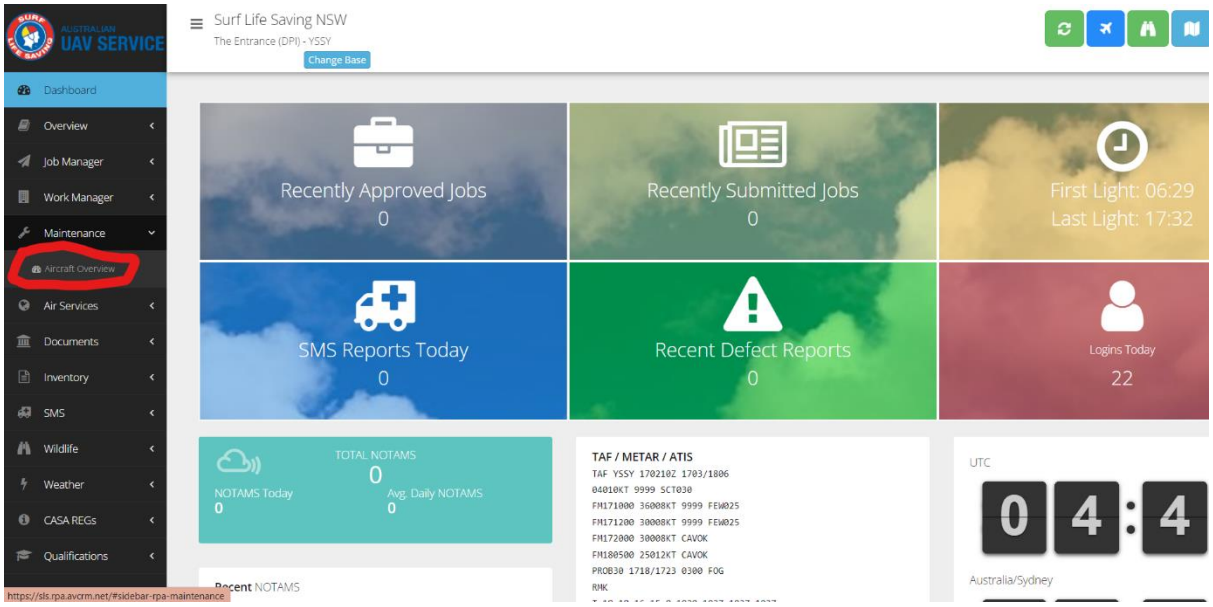
When creating your job and selecting your designated UAV, it's important to note the colour coding for the RPA Selection part of creating your AVCRM Job. This is so you can note the availability of your UAV you wish to use.

- Available - Green ✅ 🟩
    - Means that the UAV is available to use for the time that you have inputted in the estimated Job Start & Finish time.
    - You should still take precautions during your pre-flight checks and control checks to ensure there are no defects before operations.
- Booked - Blue 🟦
    - Means that someone else has also put in a job for the same time that you have inputted your estimated Job Start & Finish time.
    - There is nothing wrong with the UAV so it is still fine to use, but be considerate to the other member who may be using it at that time.
- Defect/Fault - Yellow ⚠🟧
    - There is something wrong with the UAV. This can range from minor to major faults. Potentially part or the entire UAV has been sent to be repaired.
    - You can check the Aircraft Overview page to view in detail.

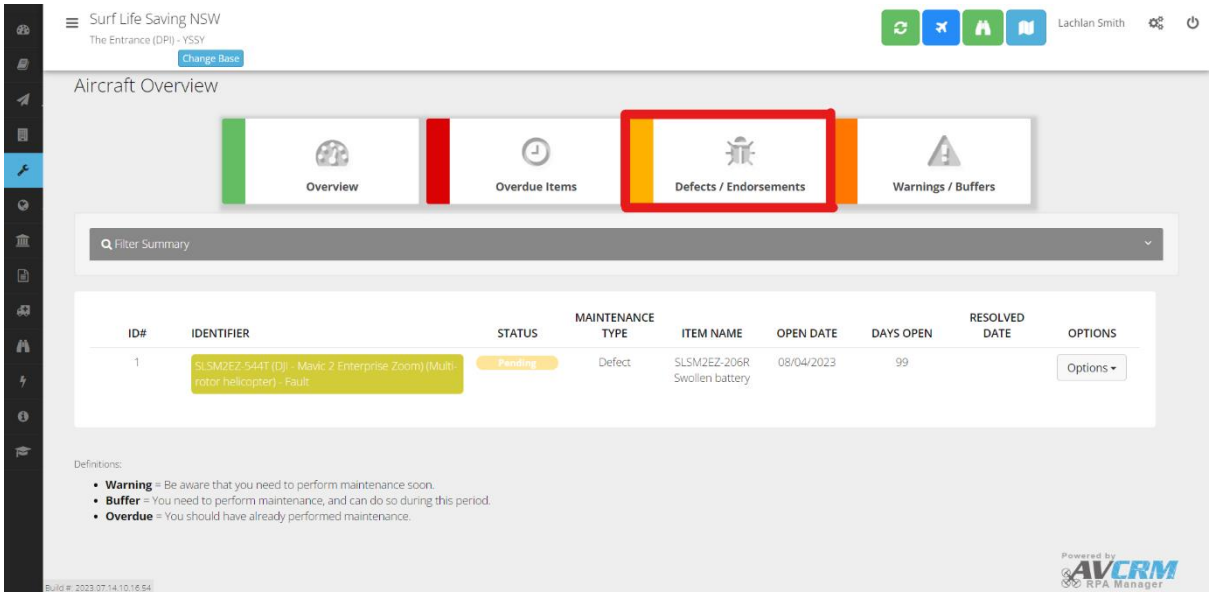
- Unserviceable - Red ❗🟥
    - The UAV has been sent into repair and the maintainer has determined that the UAV is unserviceable.
    - This will unlikely be seen on AVCRM by UAV Operators and Pilots as most Unserviceable UAVs will be removed from selection in the earliest instance.

7.2 How to review Defect Reports

All UAV Operators & UAV Pilots can review Open Defect Reports. You can do this by opening the left side menu and clicking on "Maintenance" then "Aircraft Overview".



You will then need to click on the Defects/Endorsements Button to pull all Open Defect Reports. From here you can ascertain details on the Defect and note the severity of the defect from the Defect's Item Name.



7.3 How to make a Defect Report

You can make a Defect Report after finishing a job. When Finishing a Job, the final page shows the Defect Creation Page.

1) Start by pressing the "Add Defect" Button.
2) Make sure the correct RPA (UAV) is selected.
3) Select from one of the relevant Categories.
    a) Overdue Maintenance
        i) This shouldn't be selected by UAV Operators as Maintenance is reviewed by the UAV Maintenance Controller and Location Equipment Officer(s) (LEOs).
    b) Drone (Physical)
        i) If there has been damage to the Drone's body, arms and/or legs.
    c) Battery
        i) Any defect related to the Aircraft Batteries.
    d) Crystal Sky
        i) Any defect related to the Controller's Crystal Sky (if applicable).
    e) Drone (non-physical)
        i) Software issues with the Drone components (i.e., GPS, Compass, Sensors, etc.)
    f) Other
    g) Missing Components
        i) Such as missing propellers.

h) Remote

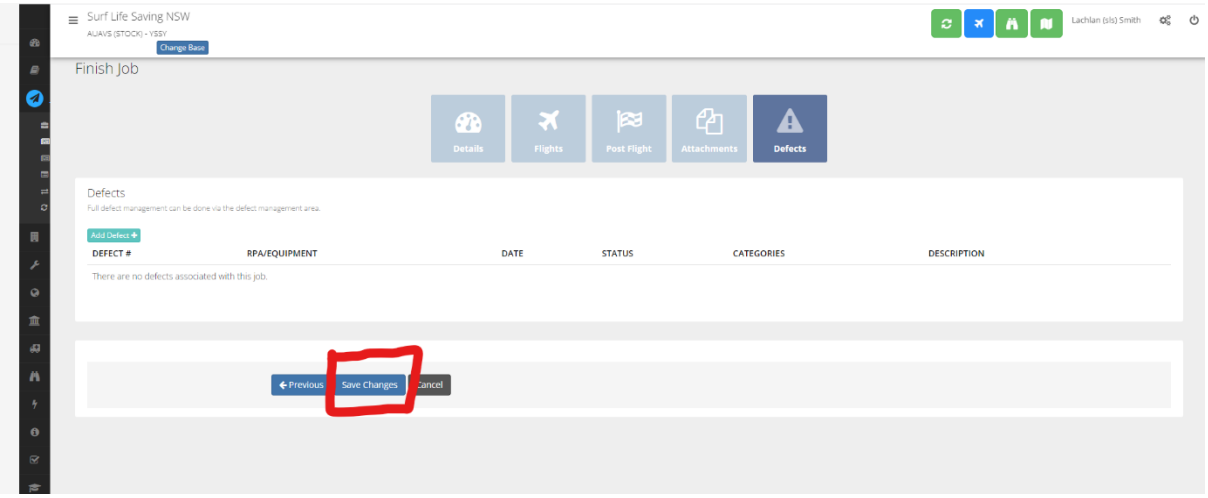
i) Any defect related to the Aircraft Remote.

4) Provide a detailed but concise description of the defect.

a) This is important as the Defect will be logged in the system and will notify the Maintenance Provider. From there the Maintenance Provider will contact the LEO or other relevant person who looks after the equipment locally to gather details regarding the Defect.

5) Save changes.

a) Do note, that once you save your job, you will not be able to adjust the Defect.



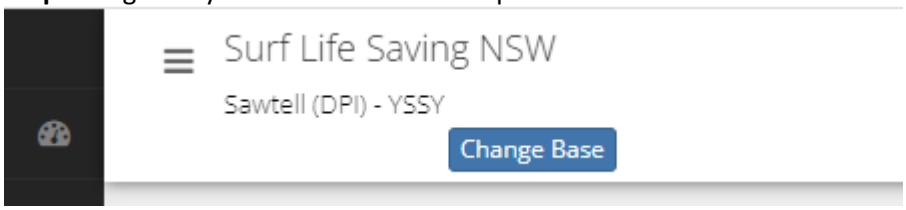
8 Creating Multi-Day Jobs on AVCRM:

This below outline explains how multi-day AVCRM jobs can be created and utilised for repetitive UAV operations. This will save you time when operating at the same location. Multi-day jobs can ease the pilot's workload by removing the need to submit a separate job for each day's flight.

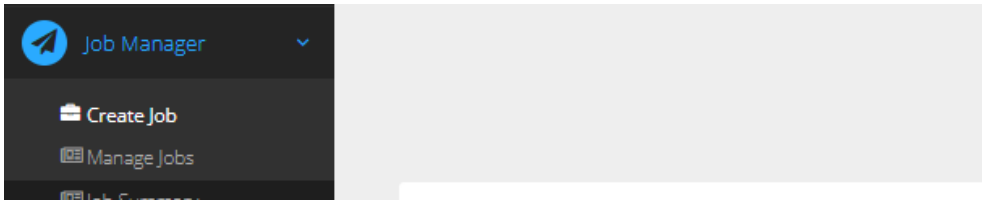
8.1 Setting up a Multi-day AVCRM Job

Multi Day Jobs can only be created for known operations in the next 30 days.

Step 1: Log in to your AVCRM account as per normal. Ensure the correct base is selected.

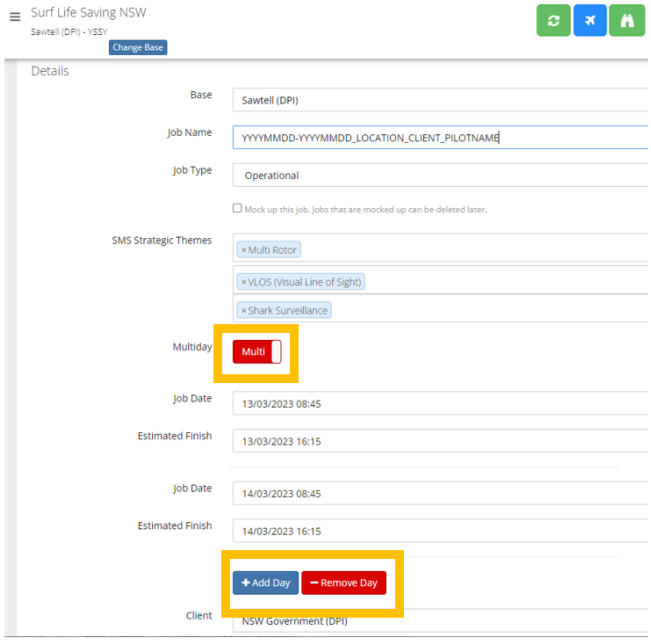


Step 2: Under the 'Job Manager' tab, select Create Job.

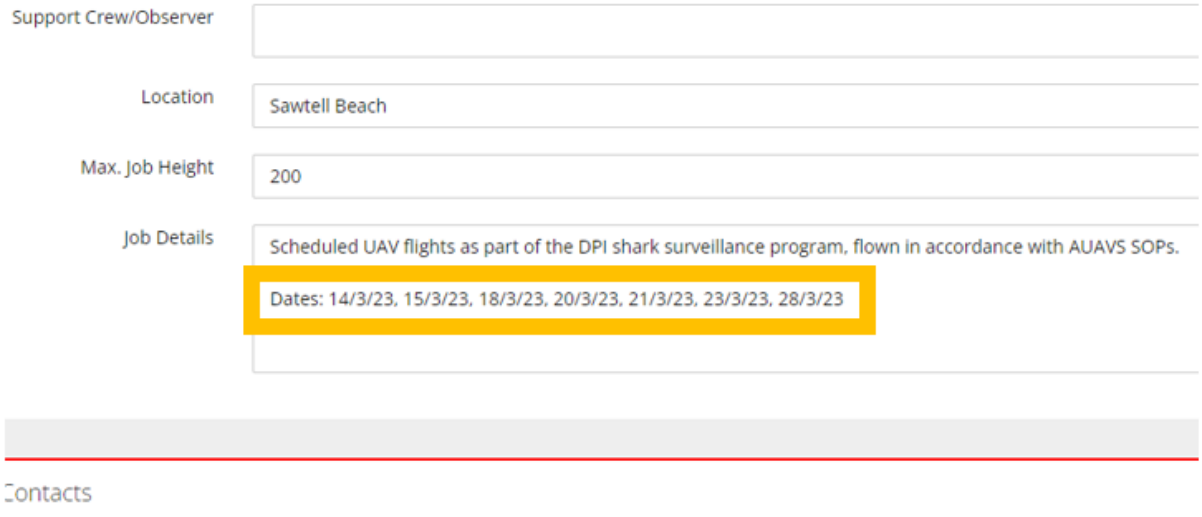


Step 3: Fill out the standard job details, while ensuring the multiday toggle switch is set to 'Multi'. For the job date, you have the option to add and remove days as necessary. **Add enough days to cover your total number of shifts** (one day per shift). Please note though, you will be unable to adjust the dates- they will simply add into the job as consecutive days.

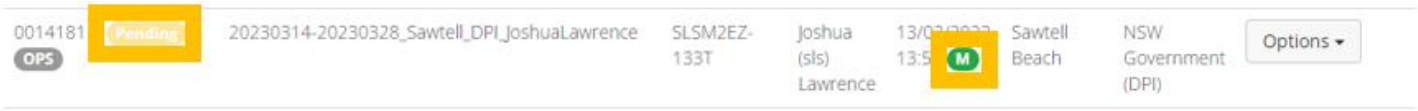
Tip: Select both aircraft at your location when creating the job in case one becomes inoperable during your job date range.



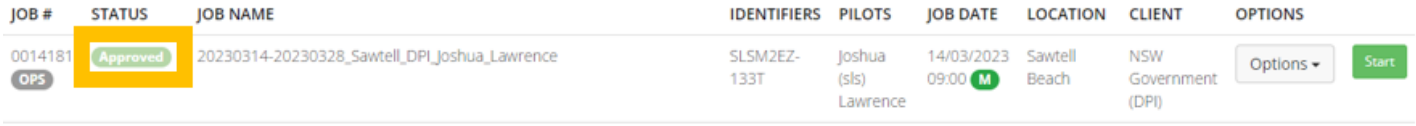
Please ensure you add your shift dates into the 'Job Details' section.



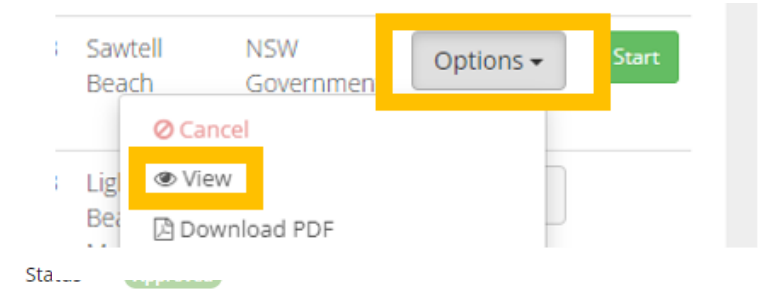
Step 4: Continue setting up your job as per normal by creating an appropriate job area, selecting the correct airspace, radio frequencies, risk assessment, and ensuring the JSA items have been selected and checked. Once complete, sign the job and click 'Save and Proceed'. The green 'M' indicates a multi-day job.



Step 5: If ReOC Job, the SBP will review the job and suggest any amendments as required. During this step the job dates will also be adjusted by the SBP to reflect the dates listed in the job details section. If it's a non ReOC job you will need to reach out to a staff member who can edit the job for you. If ReOC job it will be approved once everything has been reviewed and the job status will change to 'Approved'.



During this step, please **check to see if the dates are correct** before starting the job. Click the 'Options' dropdown menu for the job then 'View'. This will show the job information, including dates.



Multiday Yes

Day	Start(ed) At	Complete(d) At
#1	**14/03/2023 09:00**	**14/03/2023 16:00**
#2	15/03/2023 09:00	15/03/2023 16:00
#3	18/03/2023 09:00	18/03/2023 16:00
#4	20/03/2023 09:00	20/03/2023 16:00
#5	21/03/2023 09:00	21/03/2023 16:00
#6	23/03/2023 09:00	23/03/2023 16:00
#7	28/03/2023 09:00	28/03/2023 16:00

Completing and finishing the job:

Step 1: Once all details are correct, the job can be started on your first rostered shift. This process is the same as before (checking job details are correct and conducting pre-flight checklists). Once you have completed this process the job will be running.



Step 2: Conduct your UAV flights for the day, ensuring all flights are synced via the 'Flight Records' menu in DJI Pilot.

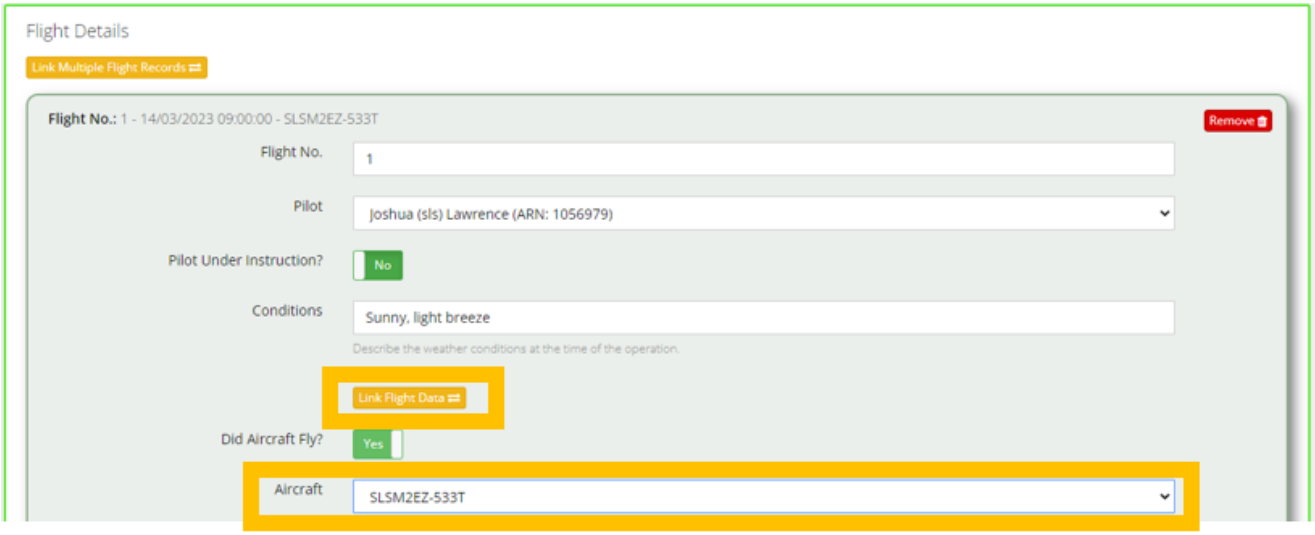
Step 3: Once all the day's flights have been imported into AVCRM you can choose to finish the job by pressing the 'Finish' button next to the job.



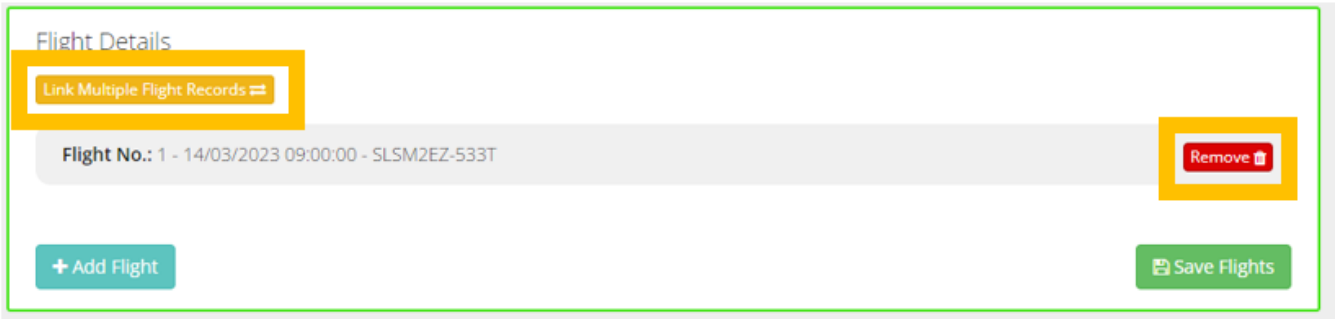
Tip: Conduct regular flight syncs through the day rather than one large sync at the conclusion of your shift. As the flight syncing process doesn't immediately import flights into AVCRM, this will minimise any potential delays waiting for flight records to appear. If you do not have any flight records appear (or they are only partially shown) you may need to try again later.

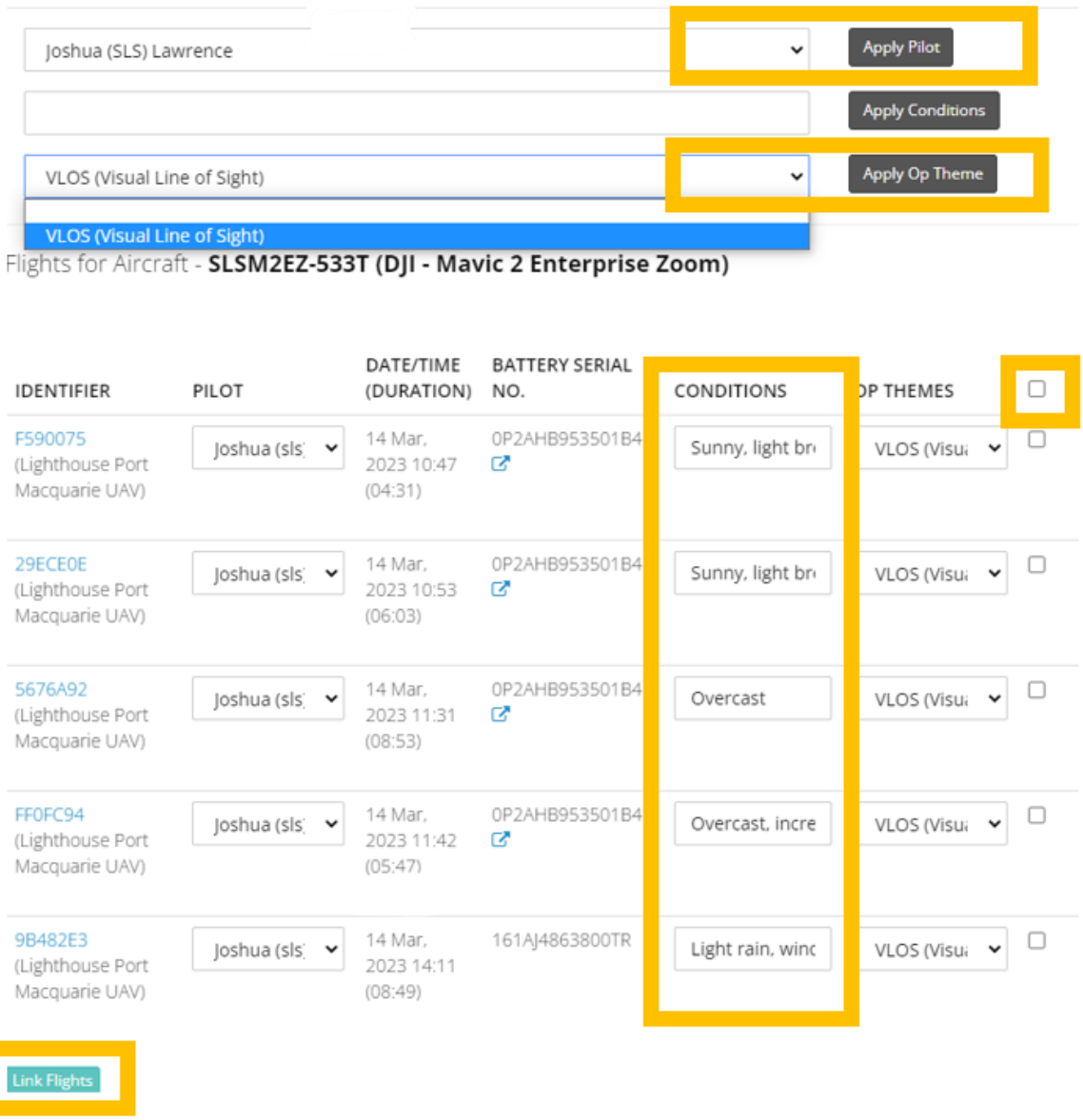
Step 4: As when finishing a job previously, add the duty time and any notes relevant for the day's operations in the 'Details' page.

Step 5: Link the flight records into the job- this is the same process as before. You can either link flight records individually to a specific flight or use the 'Link Multiple Flight Records' option to link all flight records on that date and within the job timeframe into the job. Ensure the correct aircraft is



Tip: If using the 'Link Multiple Flight Records' option, ensure you firstly remove Flight No. 1 that automatically appears as it will not have flight data populated within it.

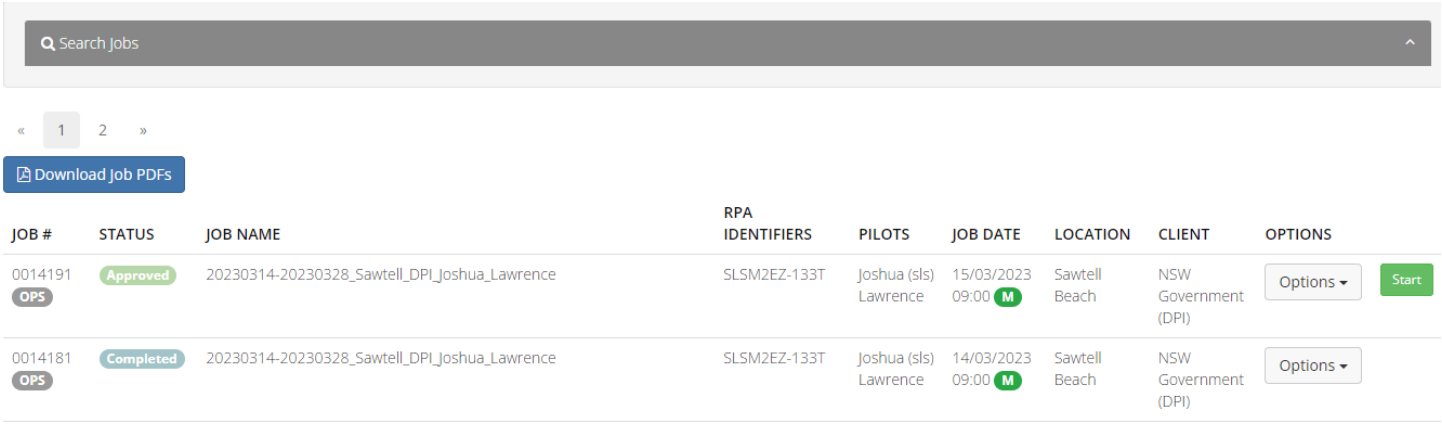



Joshua (SLS) Lawrence
Apply Pilot
Apply Conditions
VLOS (Visual Line of Sight)
Apply Op Theme
VLOS (Visual Line of Sight)
Flights for Aircraft - SLSM2EZ-533T (DJI - Mavic 2 Enterprise Zoom)
IDENTIFIER
PILOT
DATE/TIME (DURATION)
BATTERY SERIAL NO.
CONDITIONS
OP THEMES
F590075 (Lighthouse Port Macquarie UAV)
Joshua (sls
14 Mar, 2023 10:47 (04:31)
0P2AHB953501B4
Sunny, light br
VLOS (Visu
29ECE0E (Lighthouse Port Macquarie UAV)
Joshua (sls
14 Mar, 2023 10:53 (06:03)
0P2AHB953501B4
Sunny, light br
VLOS (Visu
5676A92 (Lighthouse Port Macquarie UAV)
Joshua (sls
14 Mar, 2023 11:31 (08:53)
0P2AHB953501B4
Overcast
VLOS (Visu
FF0FC94 (Lighthouse Port Macquarie UAV)
Joshua (sls
14 Mar, 2023 11:42 (05:47)
0P2AHB953501B4
Overcast, incre
VLOS (Visu
9B482E3 (Lighthouse Port Macquarie UAV)
Joshua (sls
14 Mar, 2023 14:11 (08:49)
161AJ4863800TR
Light rain, winc
VLOS (Visu
Link Flights

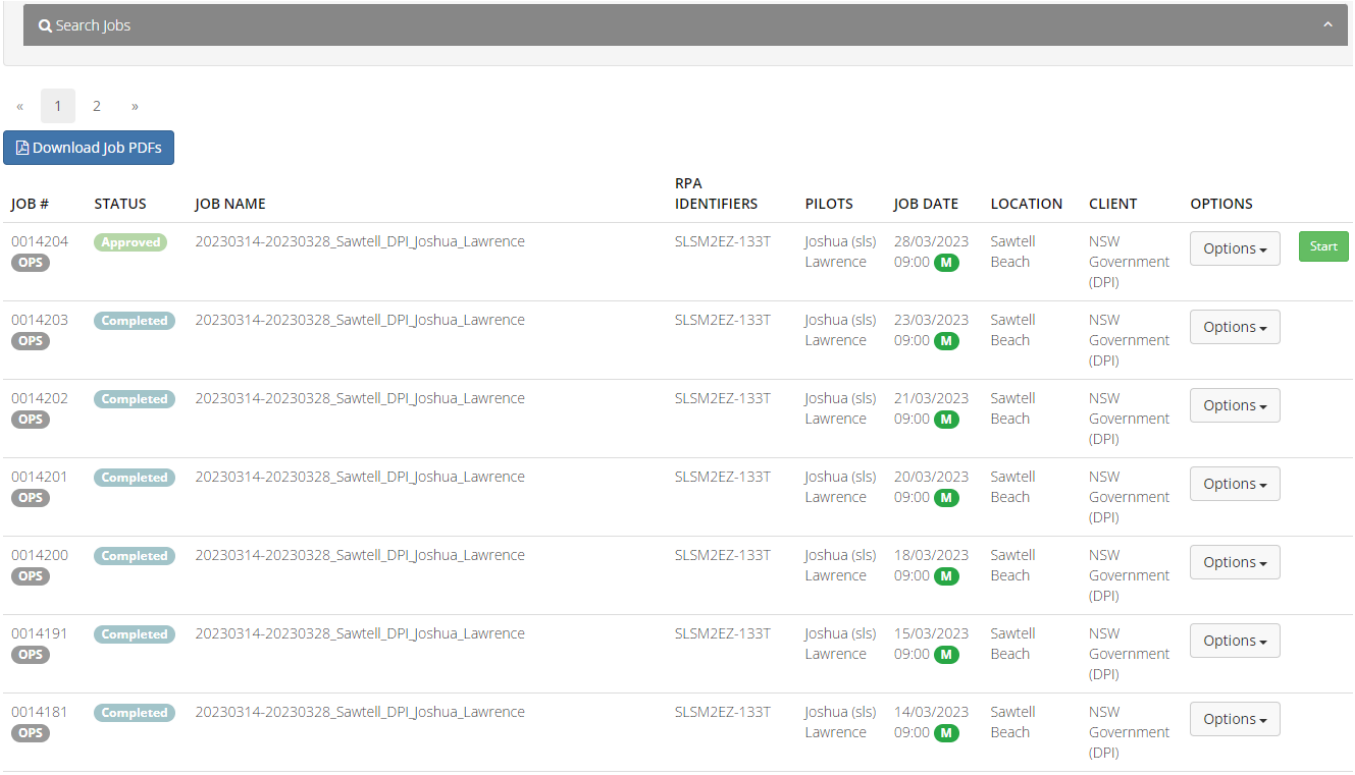
Step 6: Once flights are linked and saved, proceed to finish off the job as normal. **Please ensure any relevant wildlife sightings for each day are recorded by creating a wildlife observation**. If you are unsure of this process, please refer to the wildlife reporting guide or reach out to your relevant manager. Once complete, the AVCRM job for that date will now display as 'Completed'.

Subsequent Days' Jobs:

When you go into the 'Manage Jobs' page under 'Job Manager', your previously completed jobs will appear but there will be one new 'Approved' job at the top (this job is for your next shift at this base). A multi-day job essentially allows you to submit only one job for approval, but AVCRM will automatically create a new job with a different job number and job date for each day's operations. This process will occur for each day that was initially added to the job. All job information and specific details (radio frequencies, operations area, etc.) will remain the same. These jobs will continue to auto-create for each day until the last shift that was included in the initial job creation.



Above: after the first day's operations have been completed.



Above: completed daily jobs from this multi-day job up until the last day of operations. Note the same job title but different job numbers and job dates.

9 Pre-Operational Briefing

The following briefing is to be given by the UAV Operator/Pilot to all persons involved in the UAV operation. The UAV Operator/Pilot is also responsible to ensure the emergency contact telephone numbers are to hand.

The briefing will include an overview of the mission, all persons involved, their responsibilities and call signs, as well as identified risks and control measures.

9.1 Pre-Flight Checklist

Item	Description
Flight Planning Conducted in AVCRM:	Weather within UAV limits Airspace Approval received
Screen	Crystal Sky battery charged and operational (if applicable) Logged into DJI Pilot via the APP utilising the DJI Accounts for the base:
Controller Batteries	Charged sufficient for flight
Aircraft LiPo batteries	• Charged and condition checked • Ensure flight plan and operations ensure battery level will never go below a minimum of 30% (should be landed with no less than 30% remaining). • Low battery warning should be set to: 40% • Critical battery warning should be set to: 30%
Propellers	• Checked for: • Condition and tightness • Dents leading edges • Cracks or signs of stress • Correct Orientation • Motors Check for free movement • No abnormal movement or noises
Frame	• Visual check screws located • Check for cracks or damage • Check for security by firmly flexing
Battery	• Battery condition checked • Check for swelling • Check for leakage • Inserted Correctly
Lights	• Check for function • Check for damage
Transmitter	• Check levers and switches for security and function • Aerial check located • Aerial check for damage

	• Battery compartment checked for security and correct orientation of batteries
Gimbal	- Check for correct function - Check for damage
Obstacle Avoidance (the following is guidance for those aircraft the capacity to change these settings)	- Set to brake not avoid. - Horizontal sensing: brake 5m, alert 10m - Upward sensing: brake 5m, alert 10m - Downward sensing: brake 0.5m (goes into autoland at this height), alert 10m
Airband Radio	- Turned on - Correct frequency
Payloads	- Payloads correctly connected and tested - Shark Alarm files checked and tested for sound volume

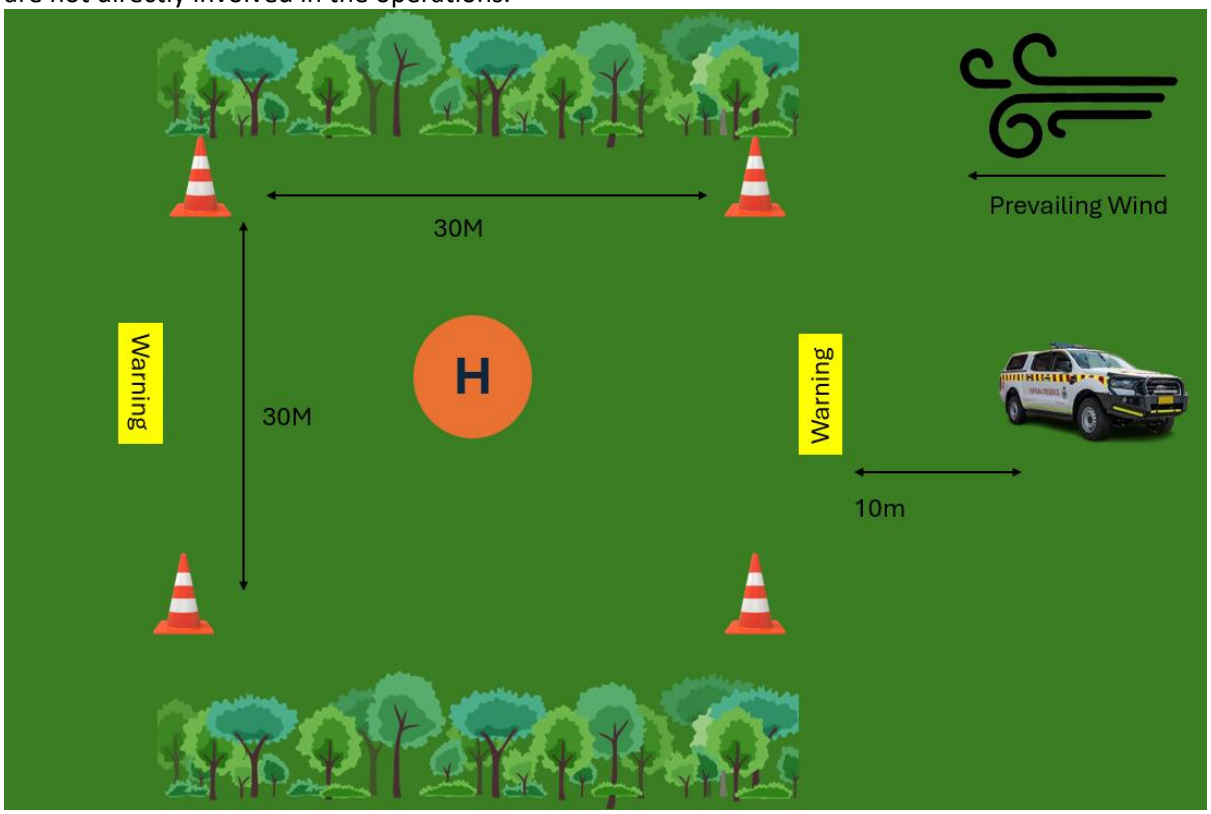
9.2 Pre-Start Checklist

Item	**Description**
Camera	Check camera FUNCTIONING
Controller	Switch ON Check battery voltage sufficient Correct MODE selected Switch positions set Fingers OFF controls
UAV	Location checked, ensure clear of populous areas
UAV Start	Insert Battery Switch ON Check ESC introduction (beeping) Power UAV (use caution, remain clear of blades)
Monitor Start	Check indication of satellite lock (minimum of 4 satellites) Check settings: - RTH height - Max distance (see below) - Max altitude (see below)
Downlink	Power video receiver and video monitor on confirm picture
Lift Off	Check controls and perform orientation familiarisation Fly mission

10 Diagram of the UAV Operations Area

10.1 Primary and Backup Landing Zones

Where possible, all staging areas must have a 30m x 30m exclusion zone set up, marked by barriers, flags or bunting, and signs to warn and restrict access to the public. If setting up such an exclusion zone is not possible, a horizontal separation of 30 meters must be maintained from all people who are not directly involved in the operations.



10.2 Proximity Restrictions

There are several proximity restrictions that are mandatory for all UAVs operated by RFS. They include:

- A flight ceiling of 120 metres (400ft) above ground level (AGL);
- Being at least 30 metres from any person not involved with UAV operations;
- At least 5.5km away from aerodromes with an operating control tower and 3 nautical mile radius of an uncontrolled aerodrome or helicopter landing site.

Noting that these proximity restrictions above are normal for most operations with RFS. These may vary depending on Location and Accreditation Privileges. Be sure to be extremely considerate of these restrictions as they will not always be the same between two locations. Be sure to use tools such as AVCRM's Airspace Calculator or ok2fly. A consequence of this may result in disciplinary action from RFS or even enforcement of regulations and legislation by CASA.

All RFS UAVs should be checked to be electronically limited (geo-fence) to a maximum horizontal distance and a maximum altitude:

All current UAV controllers display the distance from the home point, not the distance from the pilot. Pilots must ensure they have set and updated their home point at the take-off spot to maintain

accurate geofence boundaries and comply with the below distance limit from the controller. This also ensures the aircraft lands in the correct location if Return to Home or other failsafe measures are activated. Walking away from the home point will alter the actual distance from the UAV, but this change will not be reflected on the distance from home readout on the remote controller.

Make and Model	Horizontal (from Pilot) (m)	Vertical (AGL) (m)
DJI Mavic Mini	500	120
DJI Matrice 30T	1000	120

Please contact the UAV team if you find that these settings have been adjusted.

Note: Visual Line of Sight (VLOS) is self-determined by the UAV Operator/Pilot at the time of flight, the above serve as the **maximum** distances.

10.3 Take-off

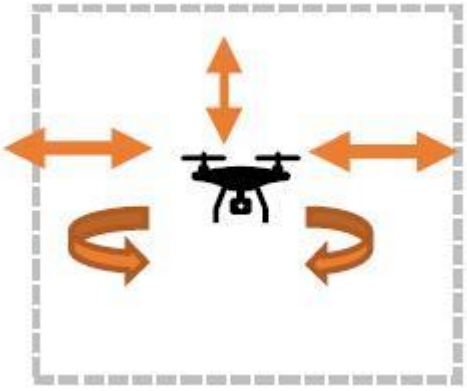
The UAV should take off and hover approximately 5 metres ("hover height") above ground level where controls and systems checks can be made prior to flight. After the checks have been made the UAV can fly towards the operational area and gradually increasing altitude. At no time should the UAV be flown over people; this includes those involved directly in the operations.

Control Check:

A Control Check is different to a Pre/Post Flight Check, in which you are checking the internal systems of the UAV by applying full power on certain components of the UAV to ensure that if at any point during the flight those components need to be utilised under stress, that they would not fail.

There is no need to do any of the checks in any order, so long that each are applied with full force on the controller and staying within your exclusion zone (which should be 30m away from any given person other than the Pilot/Operator).

- Throttle - Ascending/Descending (Left Stick Up/Down)
- Pitch - Moving Forward/Backwards (Right Stick Up/Down)
- Roll - Moving Left/Right (Right Stick Left/Right)
- Yaw - Orientate Left/Right (Left Stick Left/Right)



10.4 Landing:

The UAV must always be landed with a minimum of 30% battery remaining (should be landed with no less than 30% battery).

When returning to land the UAV should fly in within the marked operational area entering the area at a height of 5 metres slowly decreasing altitude until landed.

Best practices for UAV landing should always include reducing aircraft altitude in combination with horizontal control inputs to minimise vertical propeller wash. The UAV should be landed visually, not with reference to the display on the remote control. The UAV should be oriented pointing out, with the camera facing away from the pilot. It should be landed backwards to ensure the pilot's remote control is aligned with the UAV's heading.

10.5 Post Flight Checklist

Item	Description
Aircraft LiPo Batteries	Removed for charging
Propellers	Checked for damage
Motors	Check for free movement and for temperature changes in the motors. Temperature change may indicate a motor working harder than the others.
Frame	Check for damage Check for foreign objects
Undercarriage	Check for damage
Gimbal	Check for damage Check for foreign objects
Aircraft	Placed back into case
Sync	Sync DJI Pilot App - DJI Fly for Mavic 4 Mini Pro - Pilot 2.0 for Matrice 30T

11 UAV Cleaning

RFS has provided a cleaning kit which has all the necessary equipment to clean and maintain the UAV. Cleaning should be completed at the end of every shift, making sure to remove salt spray, sand and any other debris.

11.1.1 Cleaning Kit Contents

- Contact cleaner
- Paintbrushes
- Isopropyl alcohol bottle
- Microfibre cloths

12 Safe UAV Operations

12.1 UAV Airframe Damage

In the event of airframe damage, the UAV Operator/Pilot should make all attempts necessary to safely control and navigate the UAV to the landing zone, and if possible, land safely. If it is unable to land

safely, the UAV Operator/Pilot should control the situation and make sure that the UAV does not cause damage to people or property in the surrounding area.

12.2 Collision, Crash and Damaged Aircraft

Where a collision or crash occurs, priority shall be given to reduce any further damage or injuries to persons or property. Attempts shall be made to shut down motors utilising the combination stick command (CSC) as per manufacturer's manual via the transmitter as soon as possible, battery power shall be disconnected, and the scene preserved. Public access to the area shall be restricted until deemed safe. The CSC, as identified in the image below, must only be utilized in the event of an emergency to prevent damage, collision, or injury. Pilots/Operators must be aware of this command to ensure they do not inadvertently activate the CSC while conducting standard operations.



The UAV shall be immediately inspected for potential of a damaged battery to monitor the potential of a battery fire. The UAV will be deemed unserviceable pending inspection by the Maintenance Controller. The UAV Operator/Pilot shall notify the rostered patrol, if on-duty, and the State Operations Centre via radio. The following process should be followed:

Below are definitions of accidents, incidents, and hazards:

1) **Accident:** an accident is defined by the ATSB and ICAO as: an occurrence involving an aircraft where.
   a. a person dies or suffers serious injury.
   b. the aircraft is destroyed or is seriously damaged.
   c. any property is destroyed or seriously damaged.
   d. Please note that any accident resulting in death must be immediately reported to ATSB by the organisation
2) **Incident:** An incident is defined by the ATSB and ICAO as: occurrence, other than an accident, associated with the operation of an aircraft which affects or could affect the safety of operation.
3) Please note any incident involving aircraft owned by RFS and managed by AUAVS must be reported within 72 hours to ATSB by RFS or AUAVS.
4) **Hazard:** A Hazard is any risk to staff, members of the public or operations that is not related to the operations of the aircraft.

Pilot Reporting Process:

1) Incident/Accident/Hazard Occurs
2) Reported immediately to RFS OCC
3) Pilot Actions on day:
   a. If aircraft lost or unflyable:
      i. Pilot end flying for day/stood down.
      ii. Complete steps 3C-E
   b. If aircraft serviceable

i. Pilot pause flying while completing steps 3C-E
ii. Flying should not be resumed until the flight has been reviewed by AUAVS / RFS Management

c. Sync flight records to AVCRM immediately
d. Complete AVCRM SMS Report for Accident, Incident or Hazard depending on nature of event.
e. Defect for Aircraft created as per Defect Reporting guidelines.

4) Pilot has now completed all actions, with AUAVS & RFS to initiate internal investigation and reporting process.

12.3 Lack of Airspace Separation

In the event that "lack of airspace separation" occurs i.e. where there is a risk of the uncrewed aircraft intersecting with the flight path of a crewed aircraft or other UAV, the following actions should be taken:

- Situation assessed by UAV Operator/Pilot in command;
- Altitude dropped appropriately;
- Further evasive manoeuvres conducted if necessary;
- Hold position or initiate Return to Home (RTH);
- Aircraft landed; situation re-assessed.

In this scenario it is paramount that the UAV Pilot and flight crew act diligently to increase the distance between the two aircraft. A crewed aircraft has right of way over an UAV under all circumstances.

RFS UAV operations may encounter other recreational UAV's within the operational area, use the same actions as the "Lack of Separation Scenario" to decrease the risk of collision.

13 Maintenance

13.1 Aircraft Maintenance

All UAVs in the fleet are required to undergo a minor maintenance service every 50 flight hours and a major maintenance service once per year or every 200 flight hours. Minor maintenance involves a range of visual and calibration checks conducted by the local UAV supervisor or delegate approved by the Maintenance Controller. Major service requires shipment of the asset back to preferred supplier with a replacement asset dispatch in exchange organised by the UAV Maintenance Controller.

UAVs in the fleet are replaced as required. Fleet replacement strategy will focus on a staggered approach across the State ensuring split maintenance and rollout cycles to always maintain overall capability.

13.2 Checklists

All equipment maintenance should be carried out in an orderly process and use of checklists (provided) is encouraged, ensuring that no checks are missed. A number of these checks have been detailed already and so are linked below:

1. Pre-Operational Checks
2. Post-Operational Checks
3. Defect Logging

13.3 Battery Management

13.3.1 Battery Care and Charger Use

- Only use the supplied Lithium Polymer (LiPo) chargers. Using the incorrect charger or charge rate may cause damage to batteries and is potentially a fire hazard;
- Batteries must always be supervised while on charge. It is the pilot/operator's responsibility to monitor the batteries throughout the day;
- Batteries should always be balanced charged according to their associated specifications;
- Never charge hot or warm batteries, wait until they are less than 30°C or as per manufacturer guidance;
- If a battery is noticed to be swollen either during charging or flight operations, it should be disconnected immediately and placed in a safe location for observation. Swollen batteries should be replaced immediately;
- Charge in an open area on a non-flammable surface such as concrete floor or in a fireproof charging cabinet;
- If wires are accidently shorted or connected incorrectly, the battery should be disconnected immediately and placed in a safe location for observation for 15 minutes;
- For batteries that do not have overcharge protection, do not over charge batteries where batteries fail to stop charging once full.

13.3.2 Battery Storage

- Batteries to be stored within the provided GPC Safety Kits at all times when not in use;
- Store batteries at room temperature between 15°C and 27°C, or as recommended by manufacturer;
- Do not expose batteries to direct sunlight for extended periods of time;
- It is possible to transport or temporarily store batteries in warm temperatures, such as that of a vehicle, however this should be limited to no more than 2 hours at a time.

13.3.3 Battery Replacement

- Battery usage should cease when a pack loses 20% of its rated capacity;
- Internal resistance of cell breaches 6 milliohm, you can check the battery levels in DJI Pilot prior to flight;
- The process for swollen or damaged batteries follows:
    - Remove from use;
    - Report within AVCRM;
    - Arrange with Maintenance Controller or local supervisor for further instructions;
    - Batteries for disposal should be discharged using the “discharge function” on the DJI charging hub. Once completed, batteries should be wrapped in a suitable LiPo bag and disposed according to local battery disposal requirements.

14 Operational Techniques

All specific operations detailed below are subject to the full restrictions detailed in previous sections of these Standard Operating Procedures - the purpose of this section is to detail specific techniques relating to types of operations.

15 Appendix 1: Feedback on RFS UAV Standard Operating Procedures

We welcome feedback from members and staff about the content, wording, and ease of use of these SOPs.

Please visit the link here to provide feedback on the SOPs.

The team will assess all feedback on the following points, and may provide feedback about further action taken as a result of feedback provision:

- Applicability in current regulations;
- Precedence shown nationally or internationally, or within current operations;
- Level of acceptable risk for members, our partners, and the organisation.

16 Appendix 2: DJI Mavic Mini 4 Pro

16.1 Pre-Flight Checklist

Item	Description
Controller batteries	Charge sufficient for flight
Aircraft LiPo batteries	Charged and condition checked
Propellers	Checked for: • Condition; • Tightness, use tool if necessary • Dents leading edges • Cracks or signs of stress • Correct orientation
Motors	Check for free movement No abnormal movement or noises
Frame	Visually check screws located Check for cracks or damage Check for security by firmly flexing
Undercarriage legs	Check for security Check for damage
Battery	Battery condition checked Check for swelling Check for leakage Inserted correctly
Lights	Check for function Check for damage
Transmitter	Check levers and switches for security and function Aerial check located Aerial check for damage Battery compartment checked for security and correct orientation of batteries
Gimbal	Check for correct function Check for damage

16.2 Pre-Start Checklist

Item	Description
Camera	Check camera on
Control TX	Switch ON Check voltage sufficient Correct model selected Switch positions set Throttle closed
RPA	Location checked, ensure clear of populous areas
RPA Start	Insert Battery Switch ON Check ESC introduction (beeping) Power RPA using CSM (use caution, remain clear of blades)
Monitor RPA	Check indication of satellite lock
Downlink	Power video receiver and video monitor on Confirm picture
Lift off	Check controls Fly mission

16.3 Post-Flight Checklist

Item	Description
Aircraft LiPo Batteries	Removed for charging
Propellers	Checked for damage
Motors	Check for free movement

Frame	Check for damage Check for foreign objects
Undercarriage	Check for damage
Gimbal	Check for damage Check for foreign objects
Aircraft	Secured for transport

16.4 Maintenance Schedule

This aircraft will be maintained IAW Manufactures conditions or 'on condition' where the manufacture does not provide maintenance instructions, recommendations or directives. The pre and post flight checklists will be performed before and after every flight and the Remote Pilot will note any possible maintenance issues and record defects as per company policy.

The Maintenance Controller will monitor the failure rate of each RPAS element 'on condition' or subject to a maintenance schedule to ensure the Maintenance Plan is effective in minimising unserviceable items and operational disruptions.

16.5 Maintenance, Operations and User Manuals & Specifications

All approved documentation for this series of aircraft can be found online or in company storage locations specified in the following table.

Aircraft Type	Title	Web Link
DJI Mavic Mini 4 Pro	User manual(s)	https://dl.djicdn.com/downloads/DJI_Mini_4_Pro/DJI_Mini_4_Pro_User_Manual_EN.pdf

16.6 Battery Management

Battery Care and Charger Use

- Only use the supplied Lithium Polymer (LiPo) chargers. Using the incorrect charger or charge rate may cause damage to batteries and is potentially a fire hazard
- Batteries should always be balanced charged according to their associated specifications
- Never charge batteries unattended
- Never charge hot or warm batteries, wait until they are less than 30°C
- If a battery is noticed to be swollen either during charging or flight operations, it should be disconnected immediately and placed in a safe location for observation. Swollen batteries should be replaced immediately
- If wires are accidently shorted or connected incorrectly, the battery should be disconnected immediately and placed in a safe location for observation for 15 minutes
- Never charge in enclosed or hot conditions

Battery Storage

- Store batteries at room temperature between 15°C and 27°C
- Do not expose batteries to direct sunlight for extended periods of time

- It is possible to transport or temporarily store batteries in warm temperatures, such as that of a vehicle, however this should be limited to no more than 2 hours at a time

Battery Replacement

- Battery usage should cease when a pack loses 20% of its rated capacity or when the internal resistance of a cell exceeds 6 milliohms. Swollen or damaged batteries should also be replaced. Additionally, any batteries involved in a crash should be monitored to determine if any damage has occurred. Batteries for disposal should be discharged using the LiPo chargers (set to discharge – 3V per cell). Once completed, batteries should be wrapped in a suitable LiPo bag and disposed according to local battery disposal requirements.

17 Appendix 3: DJI Matrice 30 Series Aircraft

17.1 Pre-Flight Checklist

Item	Description
Controller batteries	Charge sufficient for flight
Aircraft LiPo batteries	Charged and condition checked
Propellers	Checked for: • Condition; • Tightness, use tool if necessary • Dents leading edges • Cracks or signs of stress • Correct orientation
Motors	Check for free movement No abnormal movement or noises
Frame	Visually check screws located Check for cracks or damage Check for security by firmly flexing
Undercarriage legs	Check for security Check for damage
Battery	Battery condition checked Check for swelling Check for leakage Inserted correctly
Lights	Check for function Check for damage
Transmitter	Check levers and switches for security and function Aerial check located Aerial check for damage Battery compartment checked for security and correct orientation of batteries
Gimbal	Check for correct function Check for damage

17.2 Pre-Start Checklist

Item	Description
Camera	Check camera on
Control TX	Switch ON Check voltage sufficient Correct model selected Switch positions set Throttle closed
RPA	Location checked, ensure clear of populous areas
RPA Start	Insert Battery Switch ON Check ESC introduction (beeping) Power RPA using CSM (use caution, remain clear of blades)
Monitor RPA	Check indication of satellite lock
Downlink	Power video receiver and video monitor on Confirm picture
Lift off	Check controls Fly mission

17.3 Post-Flight Checklist

Item	Description
Aircraft LiPo Batteries	Removed for charging
Propellers	Checked for damage
Motors	Check for free movement
Frame	Check for damage Check for foreign objects
Undercarriage	Check for damage
Gimbal	Check for damage Check for foreign objects
Aircraft	Secured for transport

17.4 Maintenance Schedule

This aircraft will be maintained IAW Manufactures conditions or 'on condition' where the manufacture does not provide maintenance instructions, recommendations or directives. The pre and post flight checklists will be performed before and after every flight and the Remote Pilot will note any possible maintenance issues and record defects as per company policy.

The Maintenance Controller will monitor the failure rate of each RPAS element 'on condition' or subject to a maintenance schedule to ensure the Maintenance Plan is effective in minimising unserviceable items and operational disruptions.

17.5 Maintenance, Operations and User Manuals & Specifications

All approved documentation for this series of aircraft can be found online or in company storage locations specified in the following table.

Aircraft Type	Title	Web Link
DJI Matrice 30	User manual(s)	https://dl.djicdn.com/downloads/matrice-30-series/20230918/Matrice30_Series_Quick_Start_Guide_v2.0.pdf

17.6 Battery Management

Battery Care and Charger Use

- Only use the supplied Lithium Polymer (LiPo) chargers. Using the incorrect charger or charge rate may cause damage to batteries and is potentially a fire hazard
- Batteries should always be balanced charged according to their associated specifications
- Never charge batteries unattended
- Never charge hot or warm batteries, wait until they are less than 30°C
- If a battery is noticed to be swollen either during charging or flight operations, it should be disconnected immediately and placed in a safe location for observation. Swollen batteries should be replaced immediately
- If wires are accidently shorted or connected incorrectly, the battery should be disconnected immediately and placed in a safe location for observation for 15 minutes
- Never charge in enclosed or hot conditions

Battery Storage

- Store batteries at room temperature between 15°C and 27°C
- Do not expose batteries to direct sunlight for extended periods of time
- It is possible to transport or temporarily store batteries in warm temperatures, such as that of a vehicle, however this should be limited to no more than 2 hours at a time

Battery Replacement

Battery usage should cease when a pack loses 20% of its rated capacity or when the internal resistance of a cell exceeds 6 milliohms. Swollen or damaged batteries should also be replaced. Additionally, any batteries involved in a crash should be monitored to determine if any damage has occurred. Batteries for disposal should be discharged using the LiPo chargers (set to discharge – 3V per cell). Once completed, batteries should be wrapped in a suitable LiPo bag and disposed according to local battery disposal requirements.

18 Appendix 4: Battery Management and Monitoring Guide

Charging Lithium Polymer (LiPo) batteries incorrectly leads to higher probability of accidents such as fires or explosions – these events pose a high risk to people and areas surrounding the affected batteries due to a potential of violent combustion. However, when used responsibly, LiPos are only as dangerous as their users. Here is a quick guide on how to handle, charge, store, and discharge your LiPo batteries safely.

The following practices should be adopted/avoided when charging batteries:

Adopt - ✅ Safe Practices
✓ Check charging batteries frequently and maintain active supervision of the charging site whilst conducting operations
✓ Set aside and report faulty batteries to appropriate Location Equipment Officer
✓ Allow batteries to cool prior to charging

✓ Use manufacturer approved equipment to charge batteries
Avoid - ⚠️ Unsafe Practices
× Charging without supervision × Charging faulty batteries (Puffy, Punctured, Deformed, Damage) × Charging batteries with/in high temperatures (>40°C) × Charging with non-manufacturer approved equipment

18.1.1 DJI intelligent flight battery - Auto discharge

DJI batteries will automatically discharge over a set time – as per aircraft manuals this should be set to 90 days or 50 cycles, whichever comes sooner. This ensures that the battery remains stable and reduces the likelihood of faults.

18.1.2 How to safety store LiPo batteries

The storage of any batteries that aren't charging can be done so within the provided GPC Safety Case. These cases remain effective for pure storage by ensuring the batteries are kept in a cool, dry place away from direct sunlight, are shock resistant and labelled clearly.



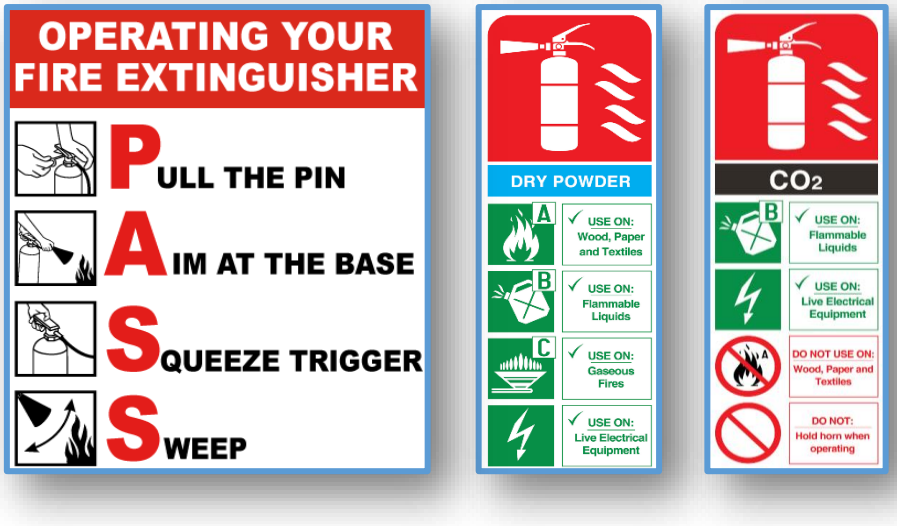
Mavic 3 Batteries in Case

18.1.3 What to do in case of a LiPo fire

In the event a LiPo fire occurs:

1. **Remain calm**
2. **Call 000**
3. **Evacuate the area**

If your location has a CO2 or Dry Powder fire extinguisher and if safe to do so, attempt the following method to extinguish the fire:



18.1.4

Never extinguish LiPo Batteries with water

Once the fire has cleared and has been deemed safe – notify your manager.

18.1.5 Disposing faulty batteries – Location Equipment Officer with approval

For Location Equipment Officers or authorized members who have faulty batteries at their assigned location please take the following steps when a faulty battery has been found.

Instructions

1. Set affected battery/s aside and inform location chat (if not already complete)

2. Create a **single** defect report in AVCRM for all affected batteries that includes **both**:
    a. asset number/s
    b. Serial number/s

3. Wait for Maintenance Controller response (Email)
    a. If warranty is approved: D1 will arrange a pickup
    b. If warranty is denied: D1 will confirm approval to dispose the battery
        - Battery disposal locations can be found here: https://bcycle.com.au/

19 Appendix 5: RPAS - Job Planning & Approvals Workflow

Job planning and approvals are essential in conducting safe operations using remotely piloted aircraft systems (RPAS). **All operations involving RFS RPAS must** have prior planning, risk assessment and approval prior to usage of these systems.

AVCRM is a system which is used to track all aspects of RPAS operations within RFS, it can be accessed using your account at:

https://nswrfs.rpa.avcrm.net

Should you have any issues with AVCRM or require advice or technical support with your RPAS or operation, please contact the AUAVS Air Desk Operator via:

- Urgent: **02 9457 1027**
- Non-Urgent: **NSWRFS@surflifesaving.com.au**

19.1 Pre-flight Workflow

19.1.1 CREATE A 'JOB' IN AVCRM

To create a job in AVCRM first navigate to the 'Job Manager' dropdown on the lefthand side of the website (fig. 1.i), select 'Create Job' (fig. 1.ii). You will then be able to begin drafting your job.

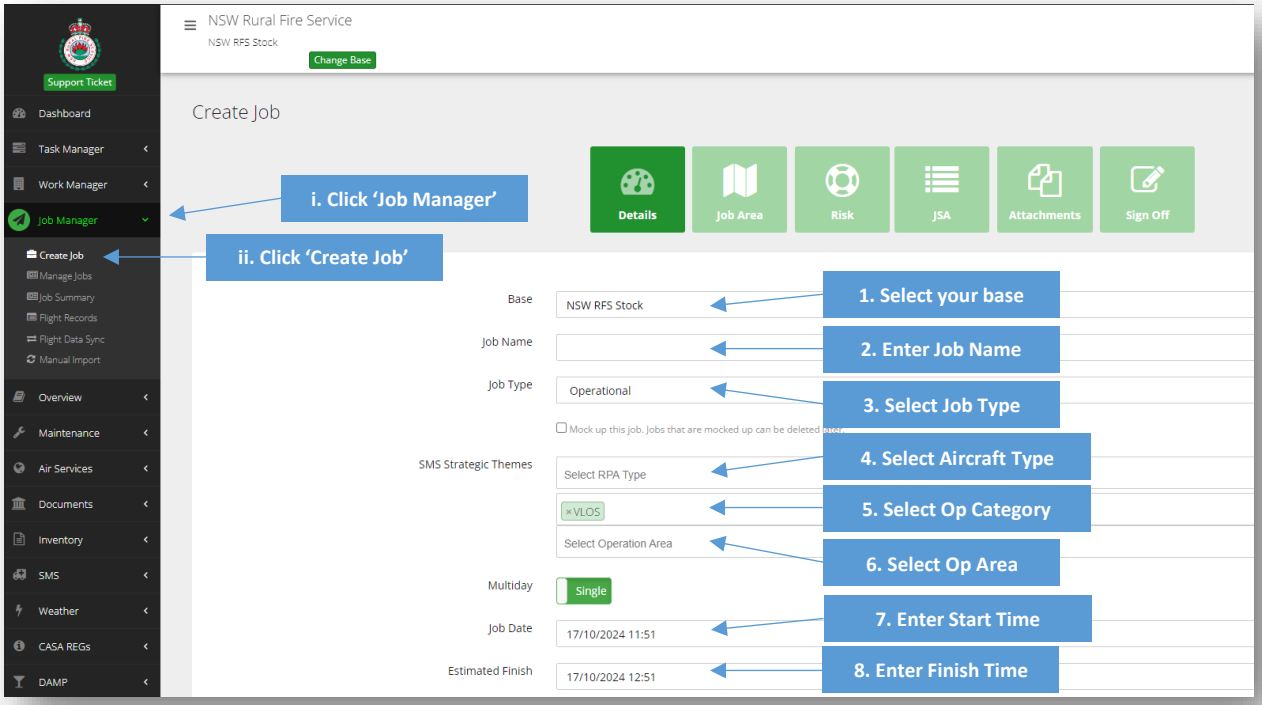


Figure 1 – Job Details

19.1.2 Job Details

1. Start by selecting the correct base for your operation in the 'Base' dropdown

2. Enter a job name using the following format

YYYYMMDD_LOCATION_NSWRFS_FIRSTLAST

- For location use the general area or suburb
- E.g. "20241001_TerryHills_NSWRFS _BrettHagan"

3. For any jobs that are operational responses to a live incident or formal tasking for a scheduled event i.e. hazard reduction activity, prefix the job name with the below;

- **PRIORITY_OPERATIONAL** _20241001_TerryHills_NSWRFS _BrettHagan
  For live incident response operations
- **SCHEDULED_OPERATIONAL**_20241001_TerryHills_NSWRFS _BrettHagan
  For planned operational activities in the coming days



4. Select a job type.

For any jobs that are operational responses to a live incident or formal tasking for a scheduled event i.e. hazard reduction activity;

- Select the Job Type as '*OPERATIONAL*'

Important: Please follow up a Priority_Operational job with a phone call to the AUAVS support line (02 9457 1027) to advise of submitted priority job request. This ensures an efficient approval response time of less than 1 hour.

For non-operational jobs, i.e. familiarisation/demonstration/ training flights etc:

- Select the Job Type as '*TRAINING'*

Please note: Non-operational job requests will be approved within 24 hours on a working day, so it is important to plan and submit these requests in advance.

5. Select aircraft type.

6. Select operation category (VLOS or NVLOS, EVLOS, BVLOS if trained)

7. Select operation area.

8. Enter start time (this should be the earliest time you expect to begin operating)

9. Enter finish time (this should be the maximum time you expect to be operating until)
   - You are not permitted to operate outside of these times – should you require to operate earlier or later than the listed times you must complete a new job requesting a new period of approval.

   - If possible, provide leeway to ensure that you incorporate the entire operation.

19.1.3

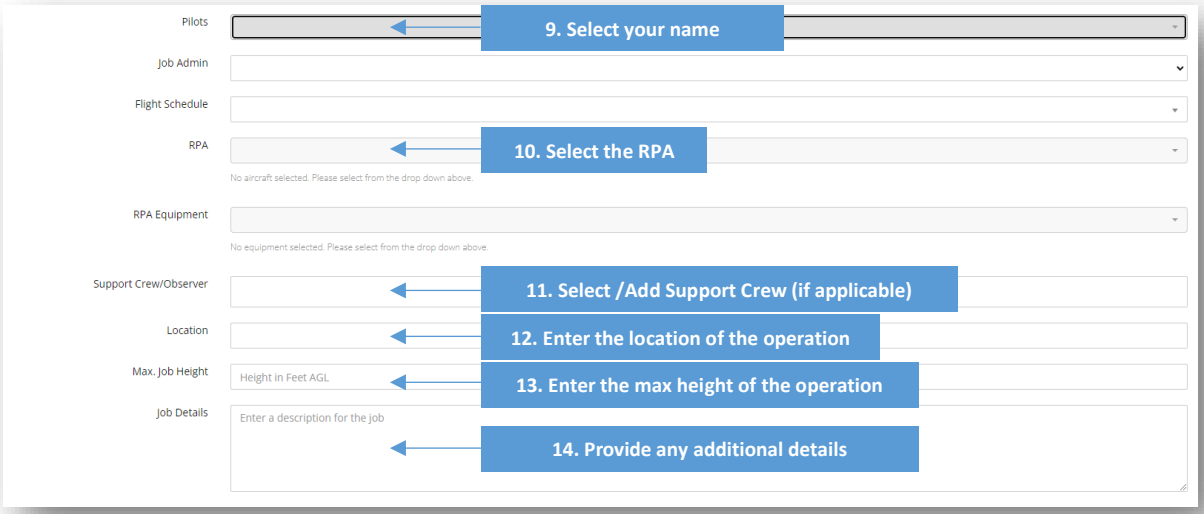


Figure 2 - Job Details Cont.

10. Select your name (including other pilots that may operate the aircraft)

11. Select the RPA you intend to fly – double check you have selected the correct aircraft.
    - Once you have selected aircraft, ensure to click 'Add Aircraft to Job'

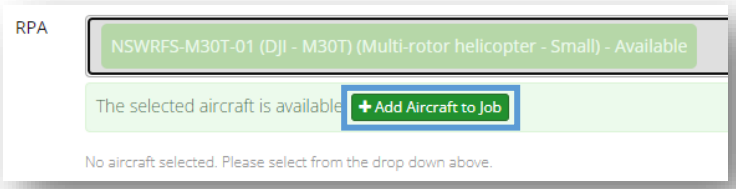


Figure 3 - RPA Selection

12. (optional) Select or add any support crew or observers.
    - Note: **this person/s cannot fly the aircraft**, if they are qualified and intend on flying the aircraft put their name in the pilot section.

13. Provide the location of the operation (Suburb, State, Post code)

14. Enter the max height of the operation

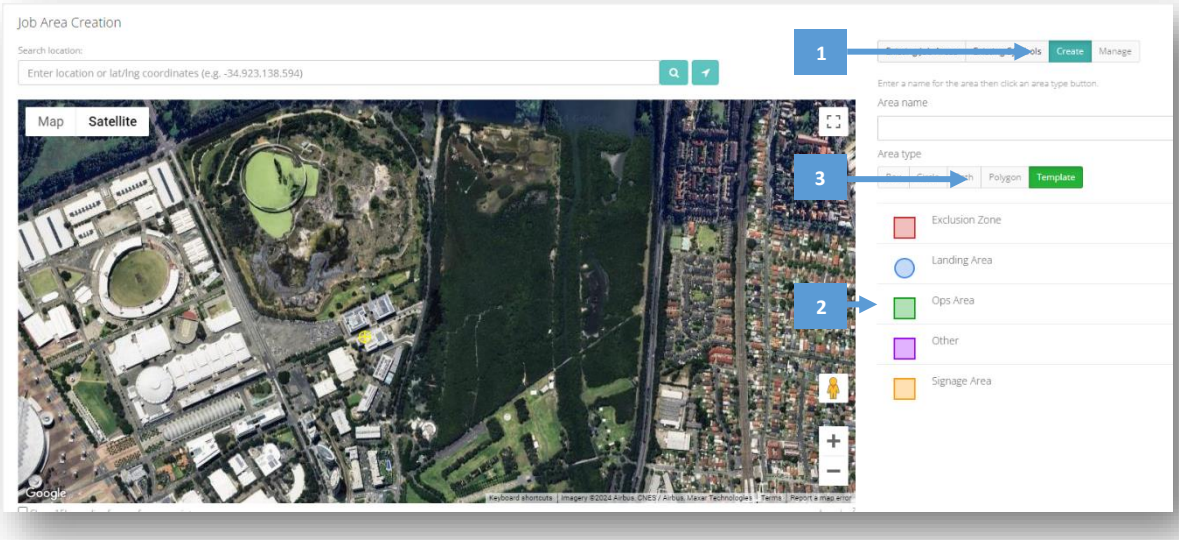
15. Provide any additional details.

19.1.4 Job Area

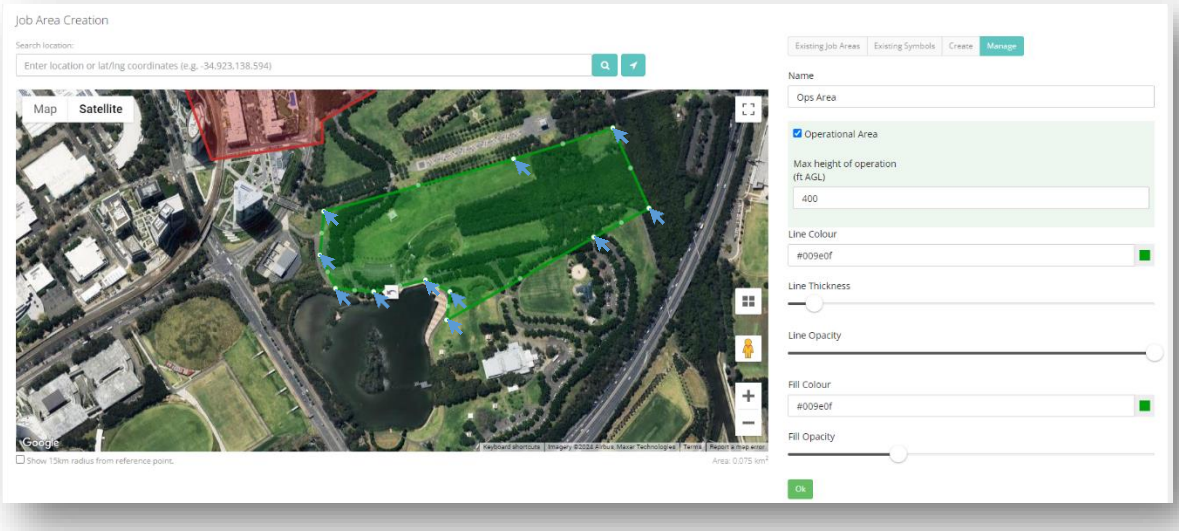
Use the job area creator to designate relevant flight areas. Each job must have at least:

- An ‘Ops Area’ – Where you will be operating
- A ‘Signage Area’ – Where your signage will be placed
- A ‘Landing Area’ – Your landing area

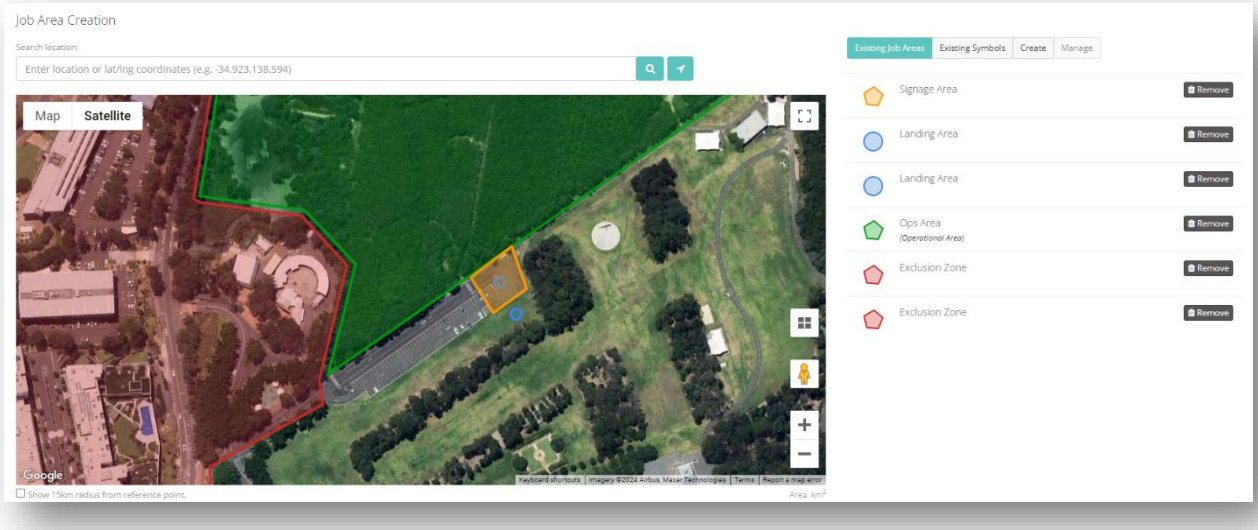
1. Select ‘create’ and select ‘Ops area’ you wish to create.



2. Use the polygon area type to designate your area, click to place a point.



3. Click ‘OK’ to save the area, then by using the same steps as above create a ‘Landing area’ and a ‘Signage area’. See the below example of a complete job area.



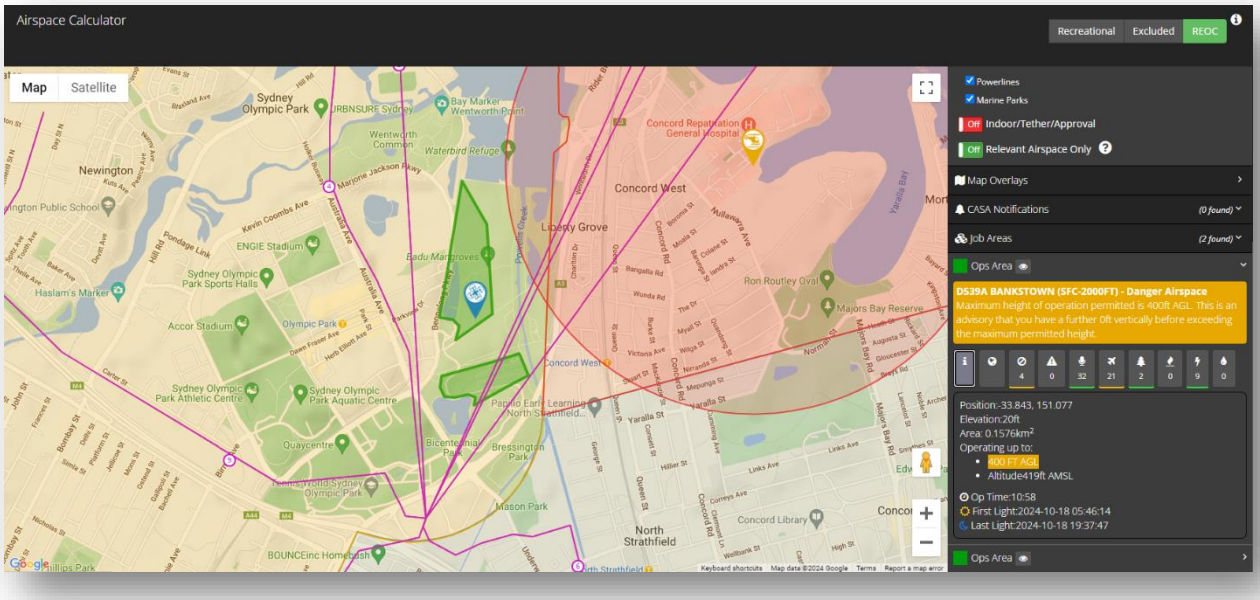
Exclusion Zones (Red area) are areas you are not allowed to fly within, or areas you do not want to fly within – for example populous areas, over roadways and aerodromes. These areas are used to prove that you are aware of these hazards and will avoid them whilst operating.

Alternate Landing Areas (Blue areas) – as you can see in the above example there are two landing areas, having two landing areas is important for contingency when your primary landing area becomes unavailable.

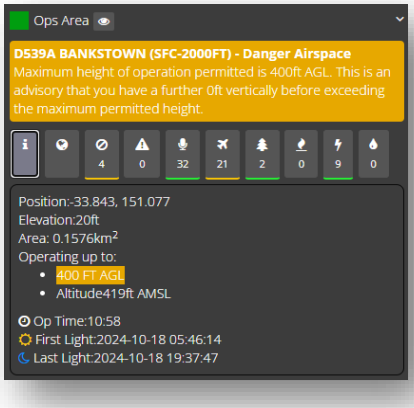
- In circumstances where flight safety is affected or unplanned landing must occur, you can land in a safe and suitable location other than what is listed in your AVCRM job area.

19.1.5 Airspace Calculator

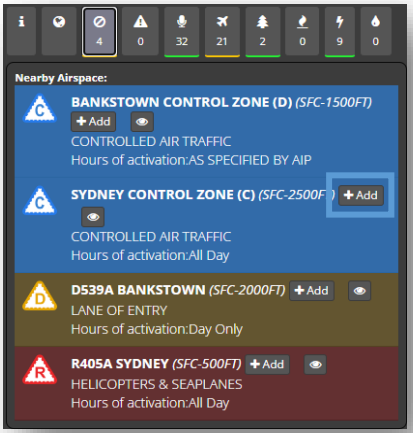
Use the airspace calculator to further inform your flight planning. The calculator will provide relevant listed airspace, radio frequencies, landmarks and hazards. Once you have completed your job area the calculator will automatically provide this information in the 'Job Areas' section.



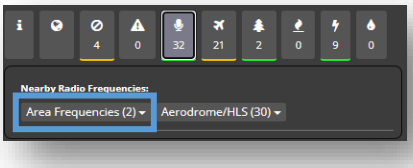
Once the calculator has updated with your 'Ops Area' (or name that you have provided), you will be provided with the following key information as well as; Nearby aerodromes (including bearing and distance), Ops area coordinates, Restricted area NOTAMs and CASA advisories. **Add relevant information to your job by clicking the 'Add button'**



General Information



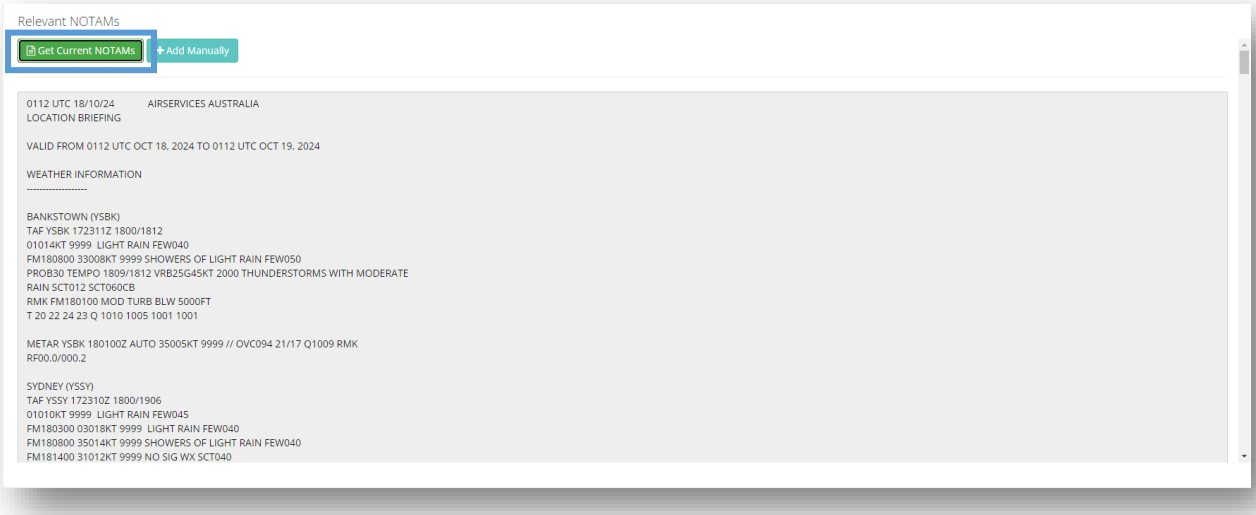
Airspace



Nearby Radio Frequencies

19.1.6 Relevant NOTAMs

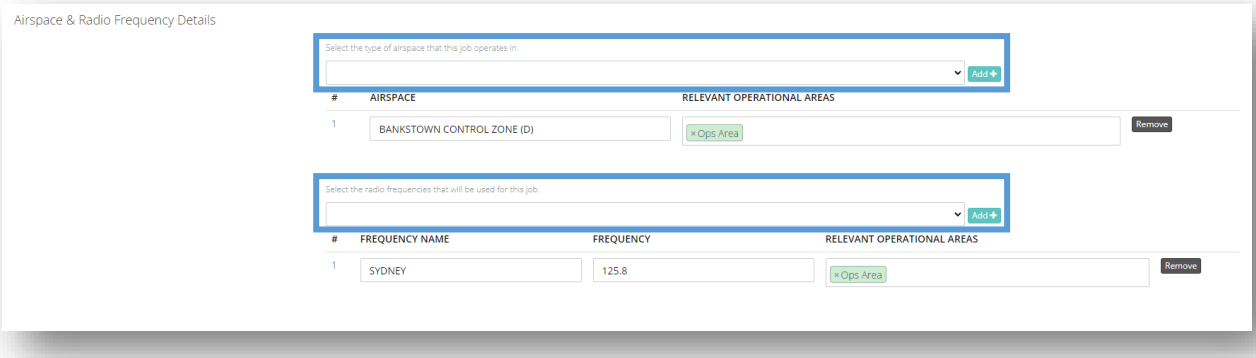
By clicking the 'Get Current NOTAMs' button, AVCRM will provide (if available) the current NOTAMs for your Ops area. Alternatively, you can manually add a NOTAM that you have acquired through Airservices by clicking the 'Add Manually' button.



Note: if you click the 'Get Current NOTAMs' button, you will be provided with NOTAM at present time and is not based on the job date or time. If your job is planned in advance check the relevant NOTAMs at the time of your operations.

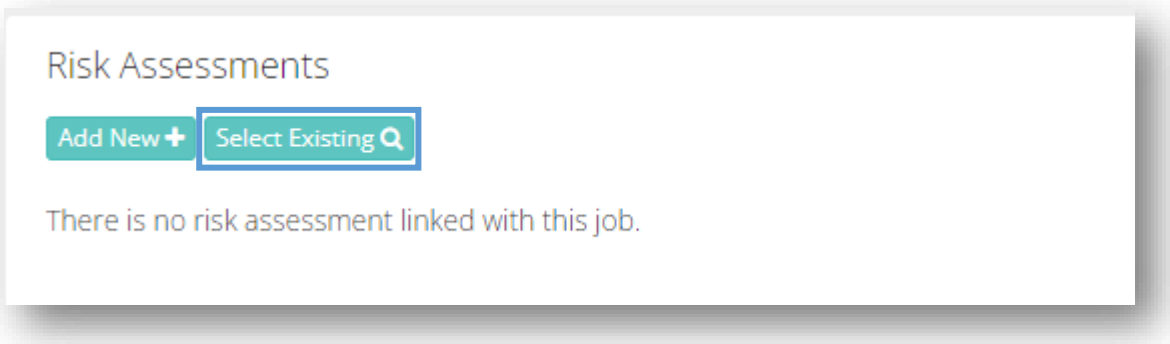
19.1.7 Airspace and Radio Frequency Details

Add relevant airspace and radio frequencies by using the airspace calculator or manually by using the dropdown box, or furthermore the manual 'Add' button.

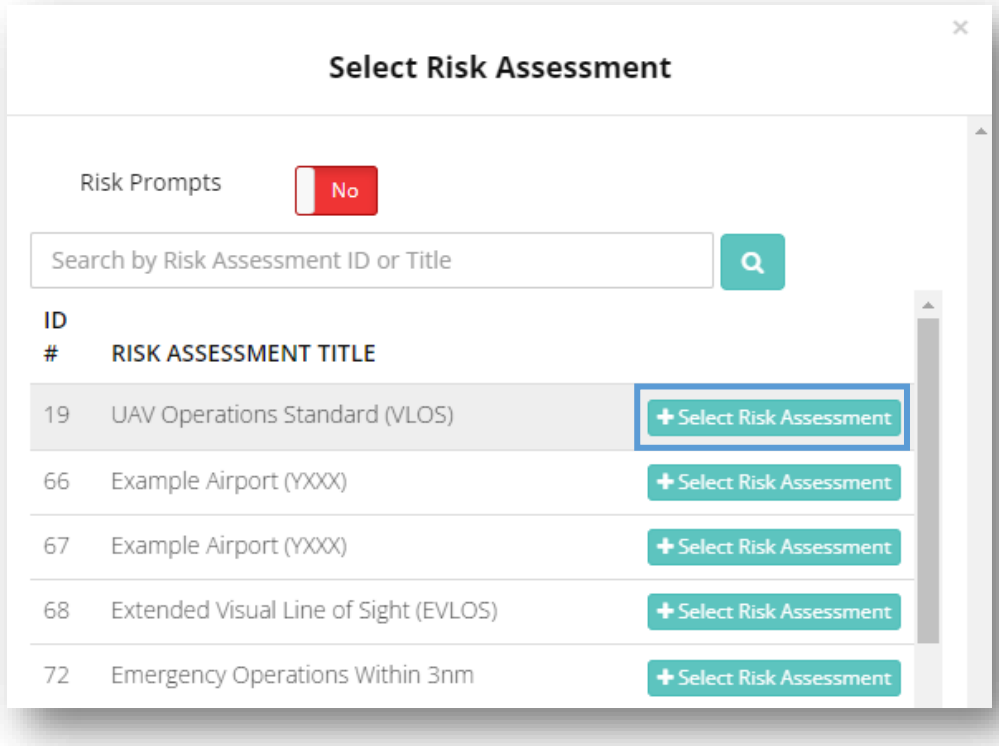


19.1.8 Risk Assessment

Click ‘Select Existing’ under the risk assessment tab and select the risk assessment relevant to your job.



19.1.9



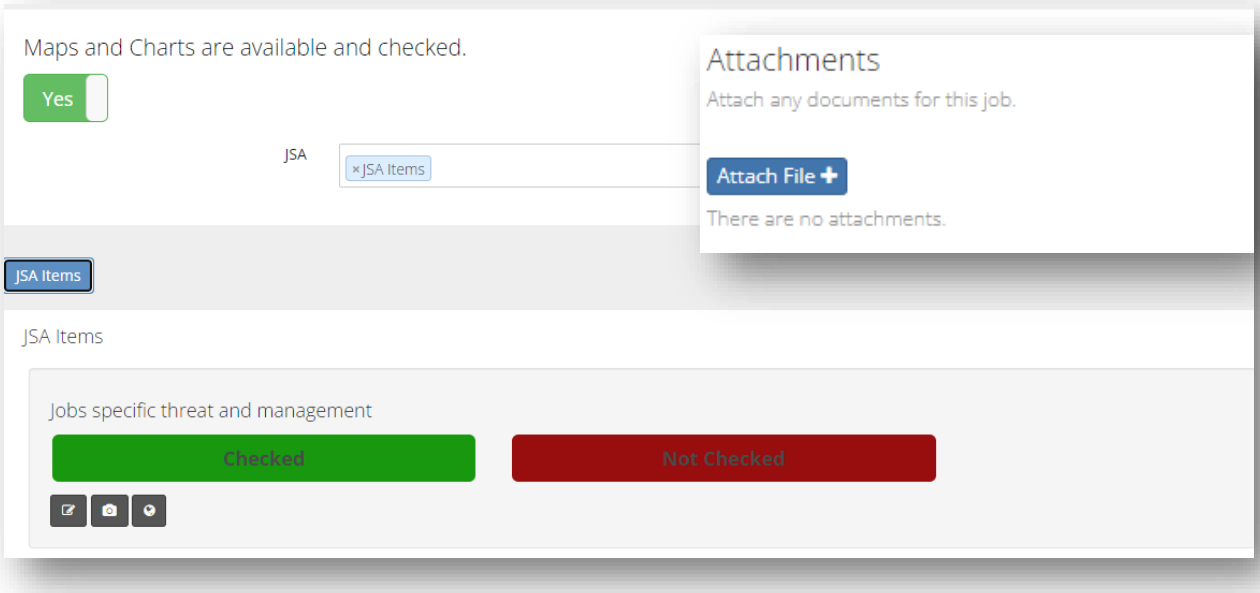
19.1.10 Job Safety Analysis (JSA)

Toggle 'Maps and Charts are available and checked' to yes – once you have checked maps and charts (you can check the maps and charts in the airspace calculator).

In the 'JSA' field – Select "JSA items", then click the 'JSA Items' button and begin to work through the checklist – selecting 'checked' or 'not checked'. You can attach a note, image or geo-tag to each section of the JSA if deemed necessary or based on preference.

You may not be able to check specific section until you are at the site – if so, you should select 'not checked' for the section and make a note that you will check prior to operation.

19.1.11



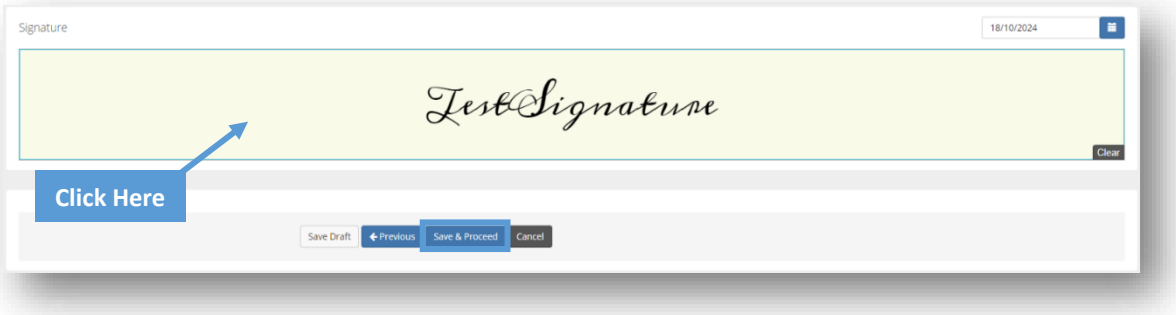
19.1.12 Attachments

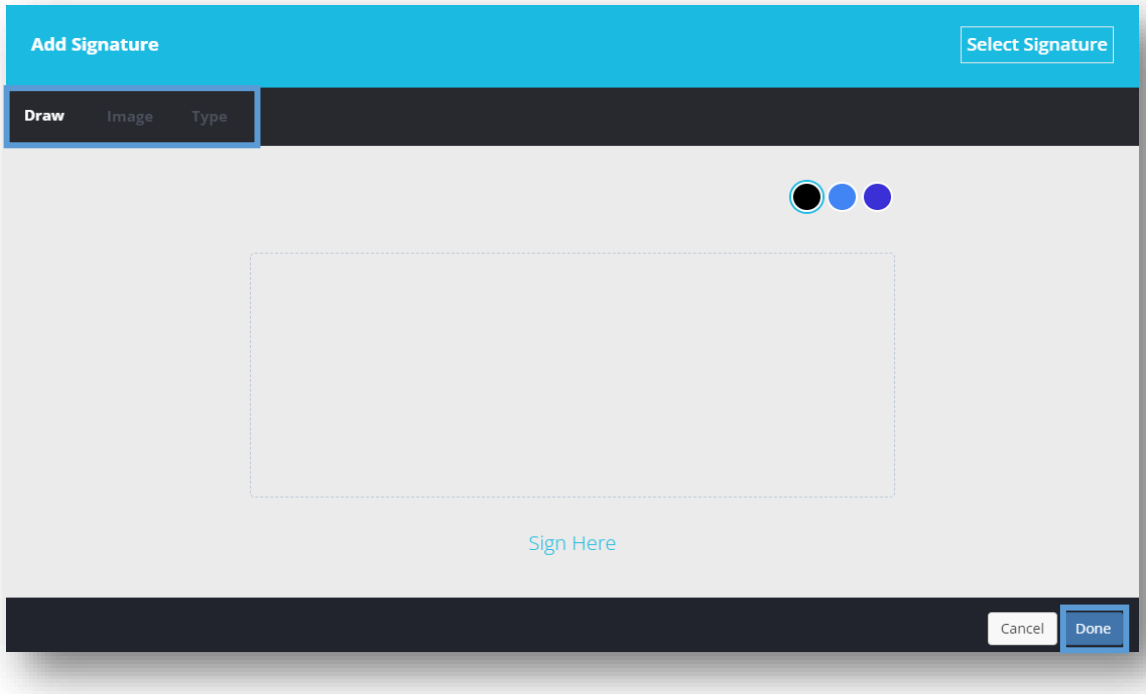
Should you have any attachments you wish to add to your job, you are able to upload them into the attachments section – e.g. if you have an agreement or approval to operate for this job, it should be uploaded here.

19.1.13 Sign-Off

Once you have completed the above and checked that all necessary items are correct and included you may now sign off your job.

Click on the yellow signing panel and you will be prompted to either 'Draw', 'upload an image of' or 'type' your signature, once completed click 'Done' and then click 'Save and Proceed'.





19.1.14 Job Submitted

You will receive an automatic response email from AVCRM confirming that the job has been submitted for review. The AUAVS Air Desk Operator will review the job within 1 hour.

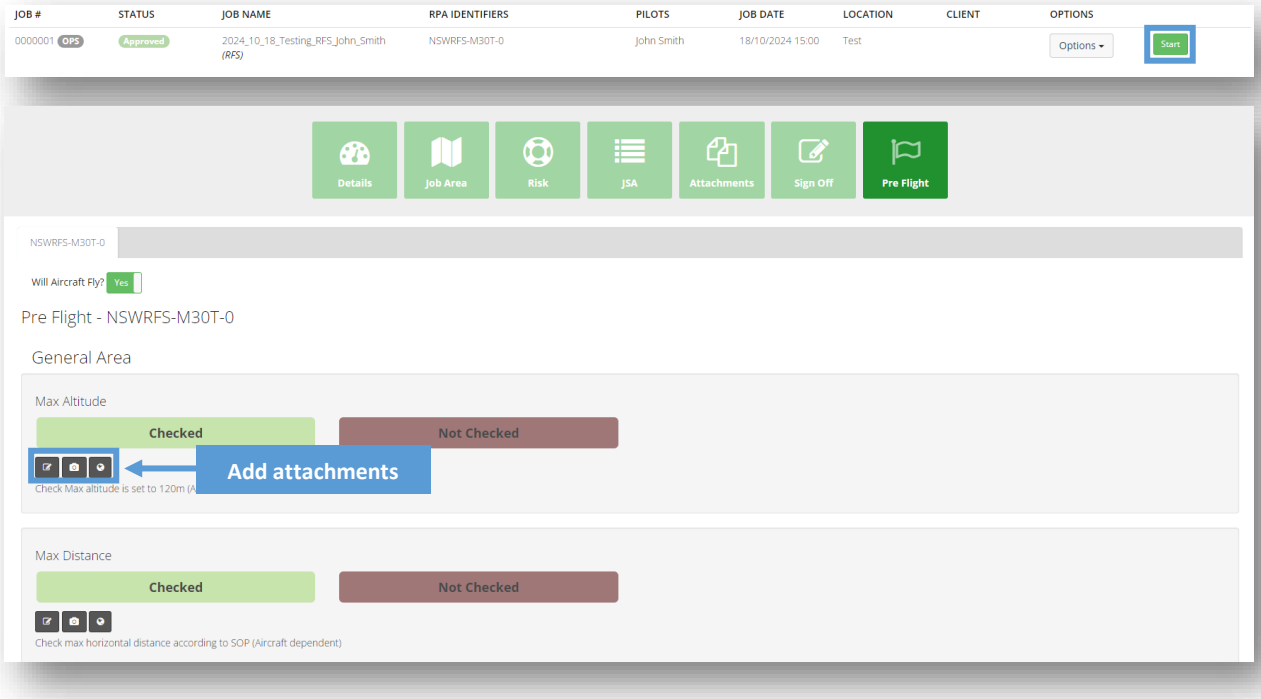
Should your job be URGENT and require approval sooner **OR** if your job has not been approved within 1 hour, please contact the duty officer on:

02 9457 1027

19.2 Post Approval Workflow

19.2.1 START YOUR 'JOB' IN AVCRM

Once approved click the green 'Start Job' button next to your job, double check that the details pertaining to your operations are complete, complete the preflight checklist and save changes. In the preflight checklist, you can add a note, image or geo-tag if deemed necessary or if preferred.



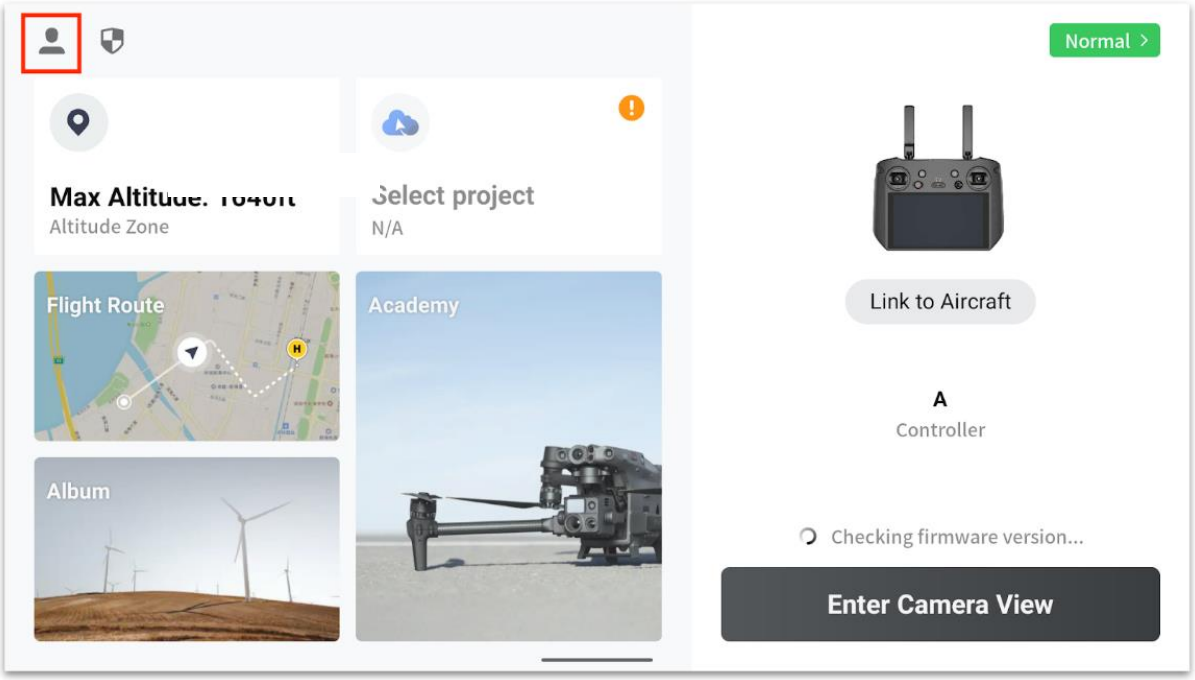
You are then ready to fly!

19.3 Post-flight Workflow

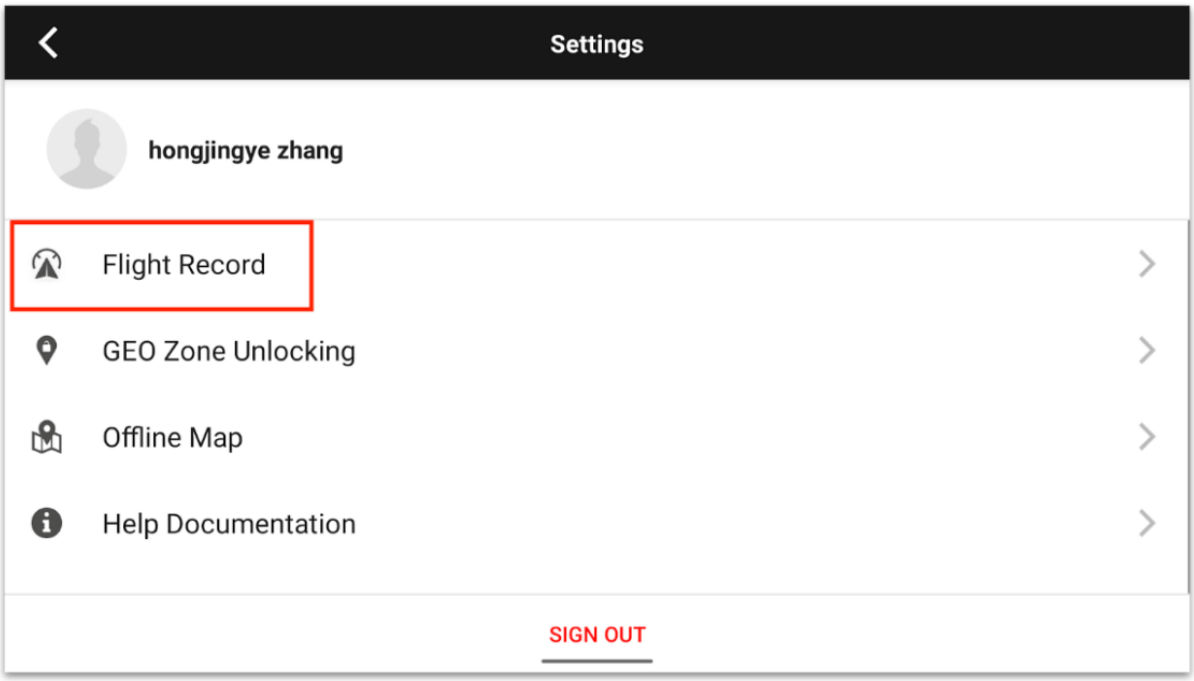
It is vital that all flight are synced and linked to the AVCRM system post each operation. This is a key step within the workflow as a pilot.

19.3.1 SYNCING FLIGHTS ON REMOTE CONTROLLER

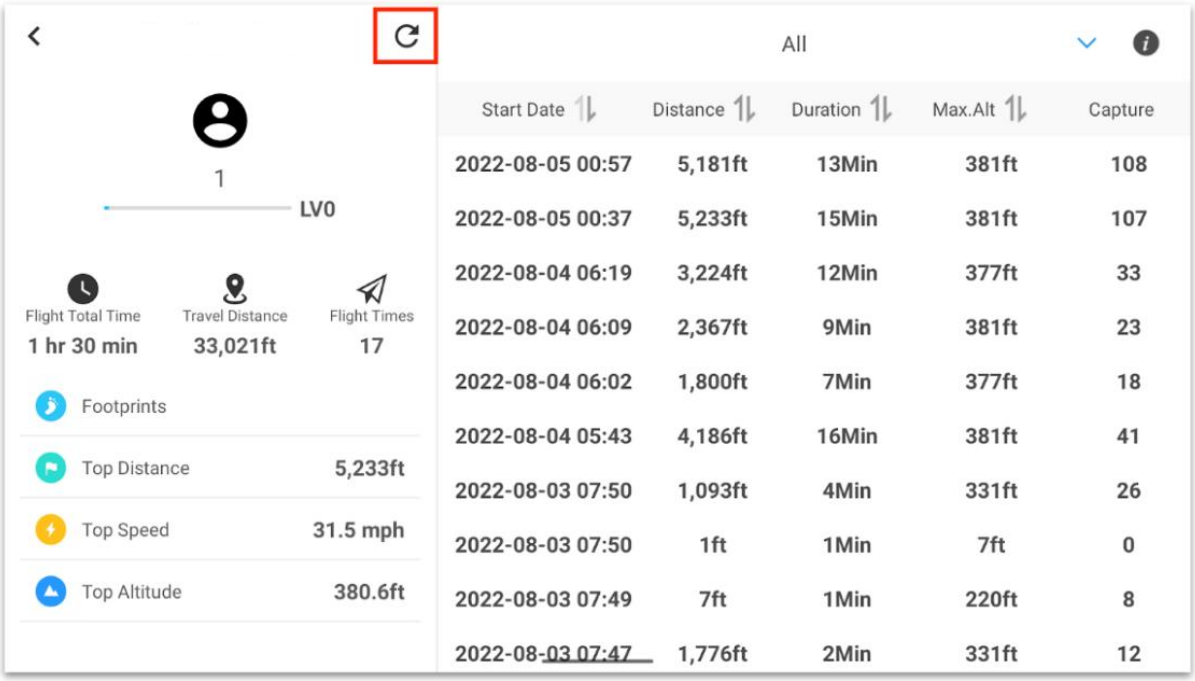
1. Open DJI Pilot and tap the User Account icon on the top left.



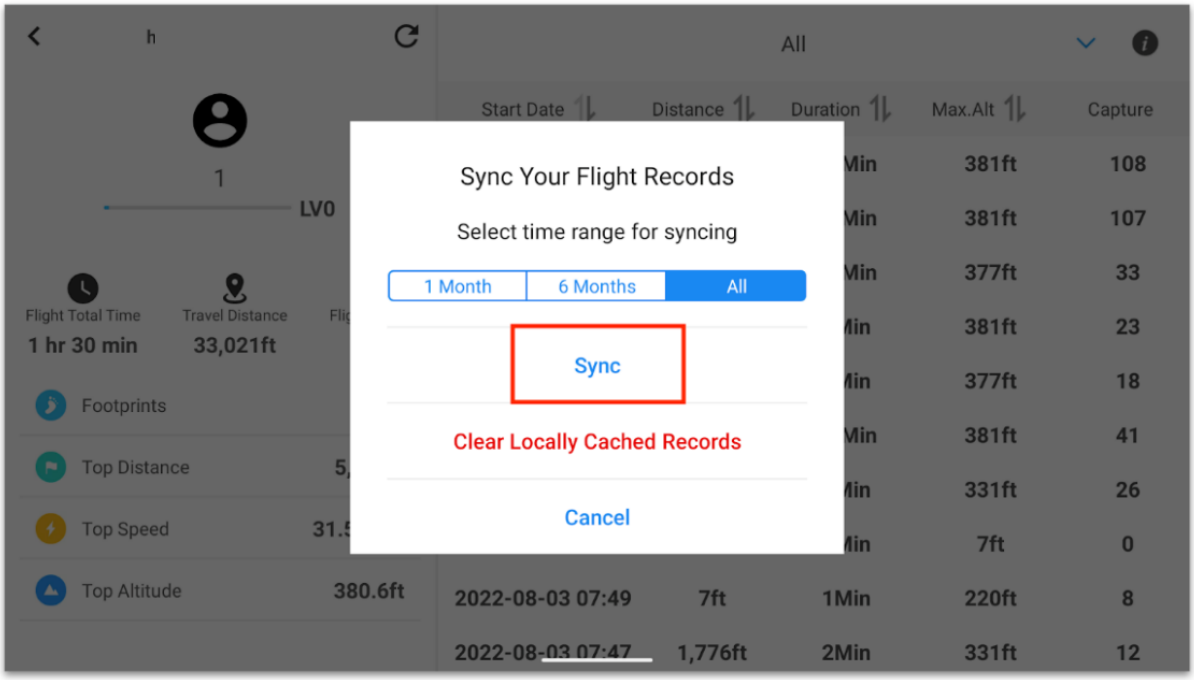
2. Tap 'Flight Record.'



3. Tap the Sync icon.

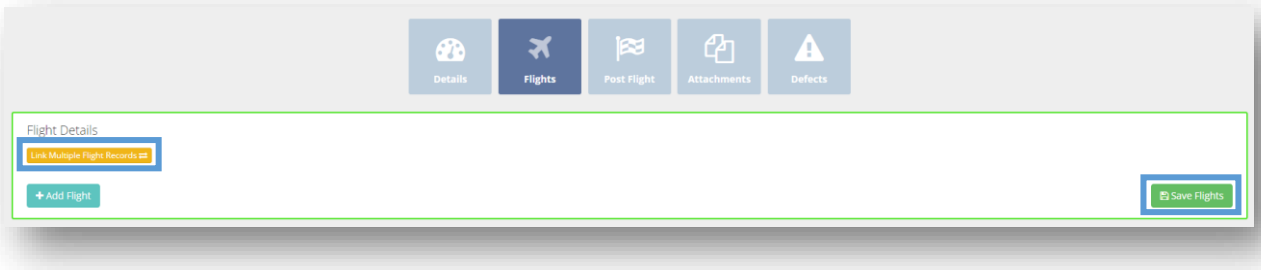


4. Tap 'Sync' to begin syncing.



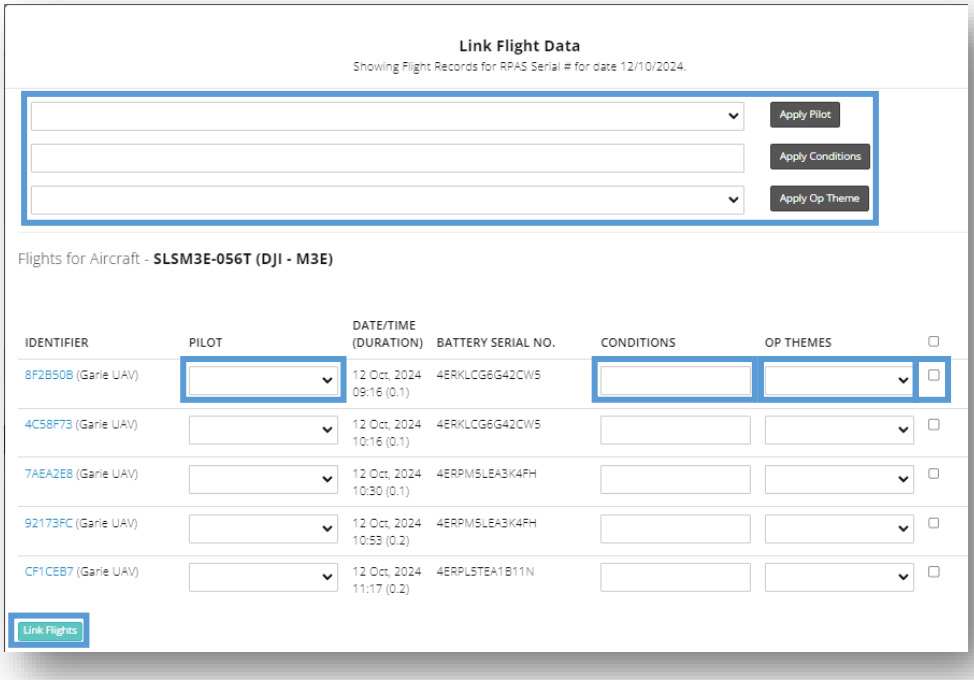
19.3.2 FINISH YOUR 'JOB' IN AVCRM

Once you have completed your operations, click the red 'Finish Job' button next to your job. In the 'Flights' tab, select the yellow 'Link Multiple Flight Records' button – you will be prompted with a 'Link Flight Data' screen.

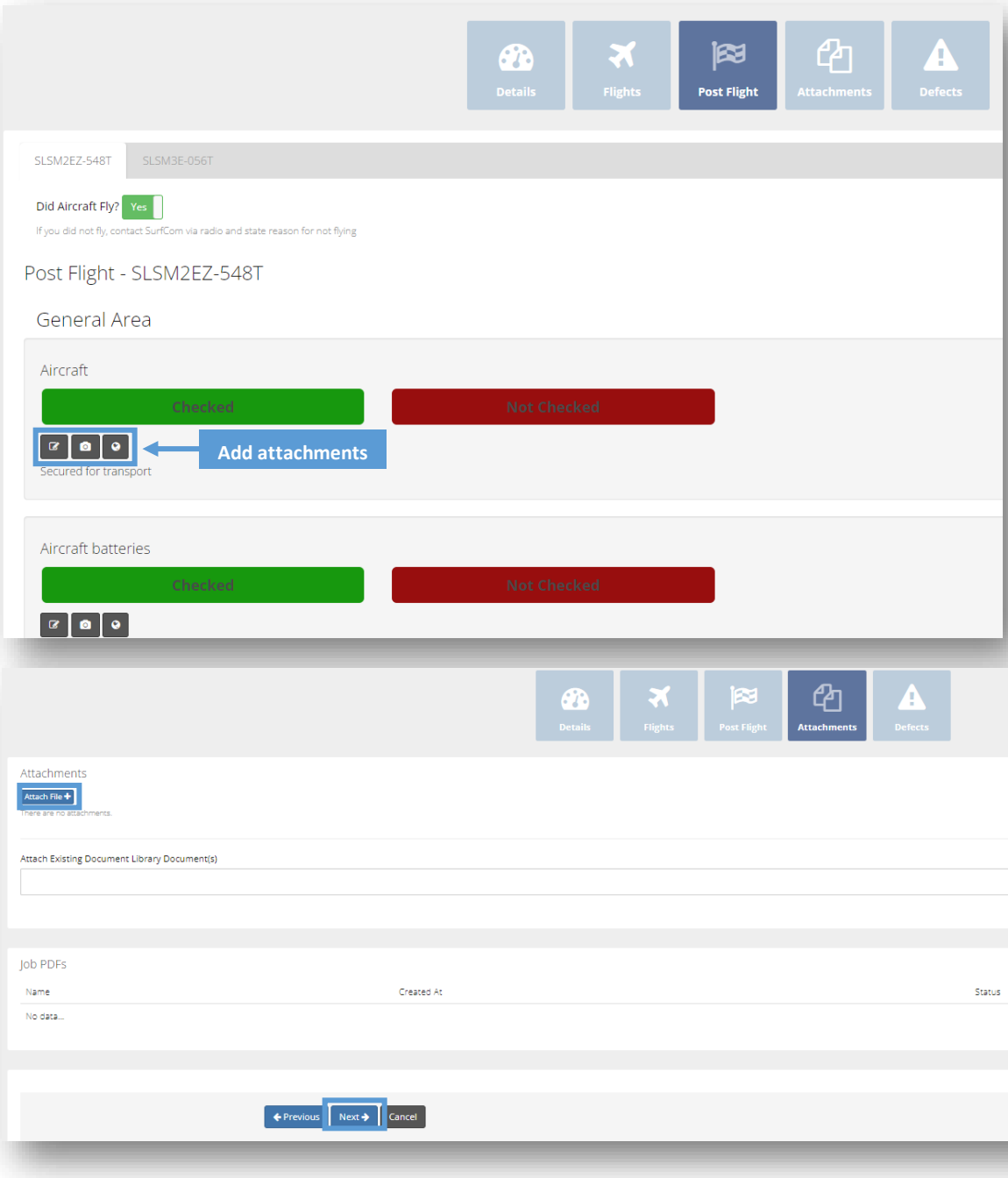


On the 'Link Flight Data' screen, you will see a list of synced flights from the aircraft and time listed on your job. For each flight select the pilot, note the conditions and select the operational theme. If all of your flights have the same information you can select them in the top 3 dropdowns for each section and then click apply for each.

Once the information has been inputted, use the check boxes on the right-hand side of each flight to select which flights you wish to link to the job – then click 'Link Flights'. The pop-up will close and then you should click 'Save Flights' and then 'Next'.



Complete the post-flight checklist and save changes. In the post-flight checklist, you can add a note, image or geo-tag if deemed necessary or if preferred - Once completed click ‘Next’.

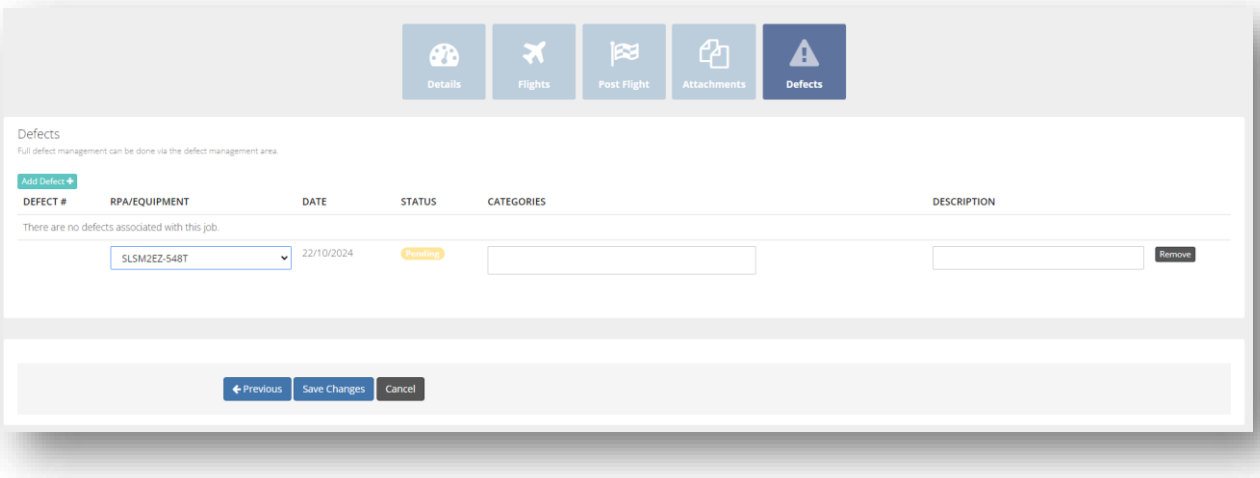


Add any relevant attachments as required or as preferred – Then click ‘Next’.

19.3.3 Defects

If you encounter any defects during the operation of the aircraft create a defect report in the final section of the job. Click ‘Add Defect’, Select the correct aircraft under ‘RPA/Equipment’, Select the correct category for the defect under ‘Categories’, and then provide a brief description of the defect under ‘Description’.

Click ‘Save Changes’ to complete the job.



Should you have any issues with AVCRM or require advice or technical support with your RPAS or operation, please contact the AUAVS Air Desk Operator via:

- Urgent: **02 9457 1027**
- Non-Urgent: NSWRFS@surflifesaving.com.au

20 Appendix 6: Spatial Data Capture/Mapping

RFS Mapping Overview

Mapping operations are carried out in both incident and non-incident settings to support data collection for features such as fire edges, hot spots, and containment lines. Coordination with data/GIS teams is essential to confirm data specifications and ensure results are provided in the required format. Mapping should be conducted under clear or consistently overcast skies, avoiding rainy conditions. The best time to minimise shadows is between 10:00 am and 2:00 pm.

Photogrammetry Capture Overview

Photogrammetry is the science of capturing and analysing detailed images to create accurate maps and models of real-world areas. Using Remotely Piloted Aircraft Systems (RPAS), this method enables efficient data collection over large or difficult-to-reach areas, producing high-quality visual outputs like **3D models** and **2D maps**. These outputs can be used for various applications, from environmental monitoring and land management to construction and urban planning.

While RPAS photogrammetry can be used to capture complex environments and challenging areas, it's essential to follow all other **Standard Operating Procedures (SOPs)** and **Civil Aviation Safety Authority (CASA) regulations**. These include maintaining **visual line of sight (VLOS)**, keeping a minimum **30-meter distance from people**, and adhering to any airspace restrictions.

Proper data management is critical to the success of any photogrammetry mission. Even with the best data capture, improper handling—like losing an SD card or formatting it incorrectly—can result in the loss of all collected data. To avoid this, it's advisable to **check the data on-site** immediately after the capture and **perform a backup** before leaving. This simple step helps ensure that valuable data is safely stored and can be used for further processing.

Types of Capture

- **Nadir Capture**:
  In a nadir capture, the RPAS camera points directly downward (90 degrees), capturing the ground in a top-down view. This method is best for flat terrain and for creating **orthomosaics**—large, detailed maps stitched together from multiple images. Nadir captures are also effective for **Digital Elevation Models (DEMs)**, which represent surface elevations without 3D features.

- **Oblique Capture**:
  Oblique captures use an angled camera, typically at 45 degrees, allowing the RPAS to capture details of vertical surfaces, such as buildings or cliffs. This method is ideal for creating **3D models** where a more comprehensive view of all surfaces is needed, enhancing realism and accuracy in complex environments.

Common Outputs from Photogrammetry

1. **Orthomosaics**:
   Orthomosaics are high-resolution 2D images that look like a single aerial photo but are stitched from multiple images. Each part of the orthomosaic is georeferenced, meaning it

accurately represents the earth's surface without distortion. This is useful for creating maps or assessing large areas in detail.

2. **3D Models**:
   By analysing multiple images of the same area from different angles, photogrammetry software can generate 3D models of landscapes, buildings, and other objects. These models represent elevation, contours, and surface details, making them valuable for simulations, visualizations, and terrain analysis.

3. **Digital Elevation Models (DEMs)**:
   DEMs represent ground elevations across an area, helping in applications like flood modelling and land-use planning. They typically exclude buildings and vegetation, showing only the bare earth's surface.

This guide will focus on capturing the data necessary for creating these outputs using the **photogrammetry capture workflow** with RPAS, following simple steps for planning, execution, and data handling. Please note that this guide does not cover steps or instructions for processing the captured data.

20.1 Photogrammetry Capture Workflow

The following instructions are based on the DJI Mavic 3 Enterprise but apply to all RPAS capable of capturing images. Software, hardware, and output may vary depending on the specific RPAS and processing software in use. Most enterprise-level RPAS offer the capability to conduct mapping missions as described below. If your RPAS does not support this feature, consider using third-party flight controller software, such as Litchi. Alternatively, you can conduct a mapping mission manually by flying and capturing images, though this method is not recommended, as it may result in lower quality and inconsistent outputs. A detailed step by step process with photos is at the end of this guide.

All steps below are for using DJI Pilot 2 flight planner. This can be found by opening the app on your DJI controller for the Mavic 3 enterprise series, M300, M350, or M30. Then select "Flight Route" to open the library of previous plans. These previous plans can be sorted by the different types of capture, by aircraft model, any associated tags, or how recently it was used/updated. In the top right, there are three buttons, "New Mission", "Search", and "Select". Under "New Mission", you have two options; "Create a Route", or "Import Route". When importing a route, it is important to use a DJI formatted KMZ/KML file. This can be either from another DJI controller, exported through Flighthub, outputted from another DJI app, or by using custom converters available online.

Planning the Capture

1. **Set Up Controller and Mapping Mission** (note below is based on DJI Flight Controller)
   - **Power On**: Turn on the DJI controller with DJI Pilot or Pilot 2 installed.
   - **Mapping Mode**: Open DJI Pilot 2 and navigate to the “Mapping Mission” mode. For non-DJI controllers or other RPAS models, settings may vary, so refer to specific device instructions. This will also work on DJI Pilot

- **Note**: To conduct mapping missions in DJI's native software, the RPAS must be part of the **DJI Enterprise series**.

Define Area of Interest

- **Objective**: Identify the specific area you want to capture. This is essential for getting useful, complete data. You will
- **Method**: Use a map or satellite image to mark the exact boundaries, covering all relevant features like buildings, trees, or natural landmarks.
- **Consider Terrain**: Take note of any height variations or structures that could affect flight safety or require additional angles.

2. **Flight Plan and Overlap Settings**
   - **Flight Pattern**: Select a "grid" or "nadir" flight pattern for general top-down views or an "oblique" pattern if capturing 3D details of structures. DJI Pilot 2 allows for both modes:
     - **Nadir (Vertical)**: Use nadir mode, with the camera pointing straight down, for large, flat areas where a 2D map or orthomosaic is needed, such as open fields, forests, or construction sites.
     - **Oblique (Angled)**: Use oblique mode, with the camera angled (often at 45 degrees), for capturing 3D models or complex structures. Oblique capture is ideal for areas with buildings, cliffs, or uneven terrain, as it gathers data from multiple angles and provides a more realistic 3D representation.
   - **Overlap**:
     - **Frontal Overlap** (along the flight path): Set to 70-80%.
     - **Side Overlap** (between flight paths): Set to 70%.
     - These settings help the software match points in each photo, creating an accurate and seamless model. For high-detail projects, consider a higher overlap as discussed in best practices.
3. **Altitude and Ground Sampling Distance (GSD)**
   - **Altitude**: Higher altitudes cover a larger area per photo, while lower altitudes capture finer detail. Choose an altitude that balances coverage with the detail required for the project.
   - **GSD** (Ground Sampling Distance): This measures the size of each pixel in the real world (e.g., 2 cm/pixel). A lower GSD gives higher detail. For example, setting a 2 cm GSD will give very fine details but may require more flights and battery changes.
   - **Note on Terrain Following**: If your RPAS has terrain-following capabilities, enabling it can help maintain a consistent GSD over uneven ground, enhancing accuracy.

- **Turn Off Date and Time Stamps**: Ensure date and time stamps are turned off in the camera settings to prevent interference with the photogrammetry software. These may need to be turned back on once operations have finished.

Conducting the Flight

1. **Pre-Flight Check**
    - **Battery and Storage**: Confirm the RPAS battery is fully charged and there is sufficient storage on the SD card.
    - **Weather and GPS Signal**: Ensure weather conditions are within operational parameters per the SOPs and the RPAS has a strong GPS signal before take-off to minimize risk.
    - **Check for Obstacles**: Perform a quick scan of the area for any obstacles, people, or unexpected terrain changes. This includes all pre-flight checks outlined in this manual such as trees, building and footpaths.
2. **Flight Execution**
    - **Launch the RPAS**: Start the flight in an open, safe area. Begin the grid pattern you selected, keeping the RPAS at a stable speed and altitude. Avoid sudden turns or altitude changes, as this can affect data quality.
    - **Monitor Status**:
        - **Battery Levels**: Keep an eye on battery levels, and plan to land prior to the battery reaching 30% to ensure safe return.
        - **GPS Signal**: Maintain a stable GPS connection, especially in areas with high structures or dense vegetation. Ensure all SOPs are followed.
    - **Safety Reminder**: Always have a visual line of sight on the RPAS, and if the GPS signal weakens or weather changes, return the RPAS to the home position.
3. **Image Capture and Quality Control**
    - **Capture Images Continuously**: The RPAS should capture images at each set interval along the grid, ensuring every frame overlaps with the previous. Avoid high speeds that could create gaps between images.
    - **Angled Shots (Optional)**: For structures with vertical surfaces, set the camera angle to capture oblique (angled) shots. This helps capture vertical details that might not be visible in straight-down (nadir) shots.

Post-Flight Data Management

1. **Data Check**

- **Onsite Review**: Before leaving the site, review a sample of images to confirm they are clear, correctly exposed, and fully cover the intended area.
- **Identify Issues**: Check for blurry or underexposed images. If any key areas are missing or unclear, consider recapturing those areas.

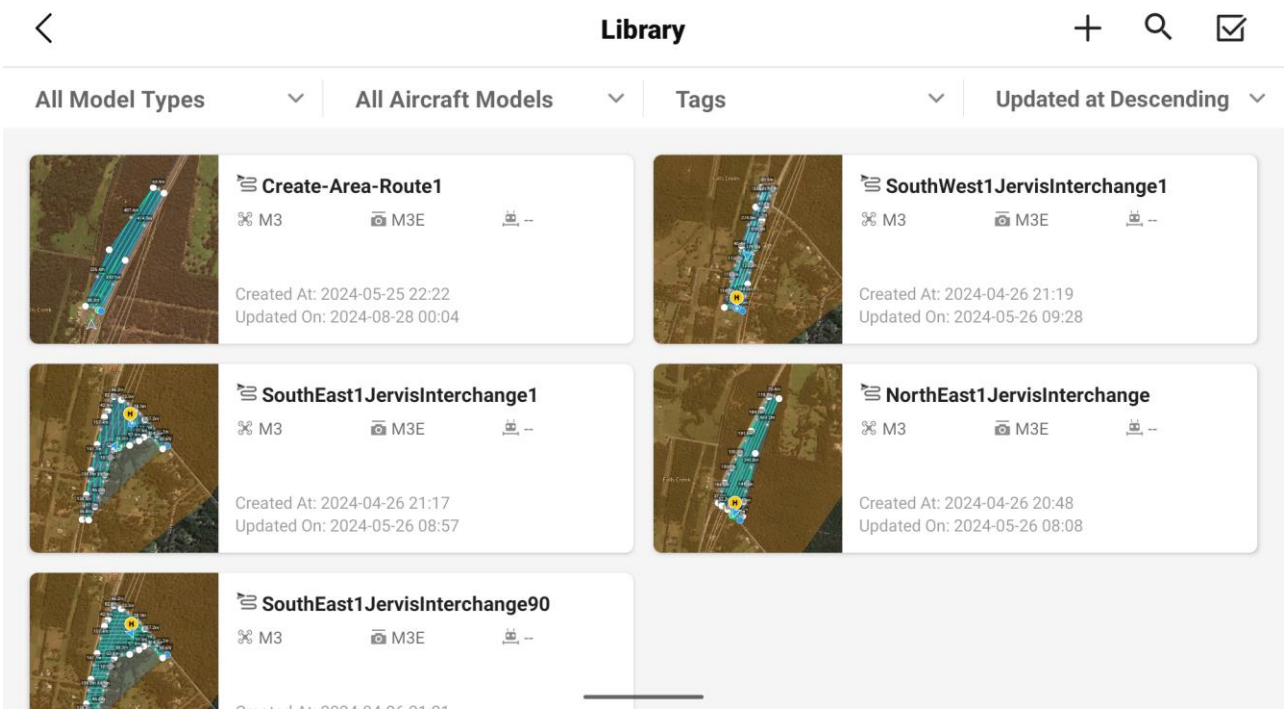
2. **Data Transfer**
   - Backup images to a secure location like a laptop or external drive.
   - Organize files by project name and date to keep data accessible and easily navigable.
3. **Prepare for Processing**
   - Store data in a format compatible with photogrammetry software for later processing.
   - Maintain organized files with any notes on GSD and overlap used, to ensure consistency and simplify processing.

DJI Pilot 2 Mission Planning Detailed Guide:

This section provides a detailed guide for mission planning using DJI Pilot 2 on DJI controllers, compatible with Mavic 3 Enterprise series, M300, M350, or M30 models.

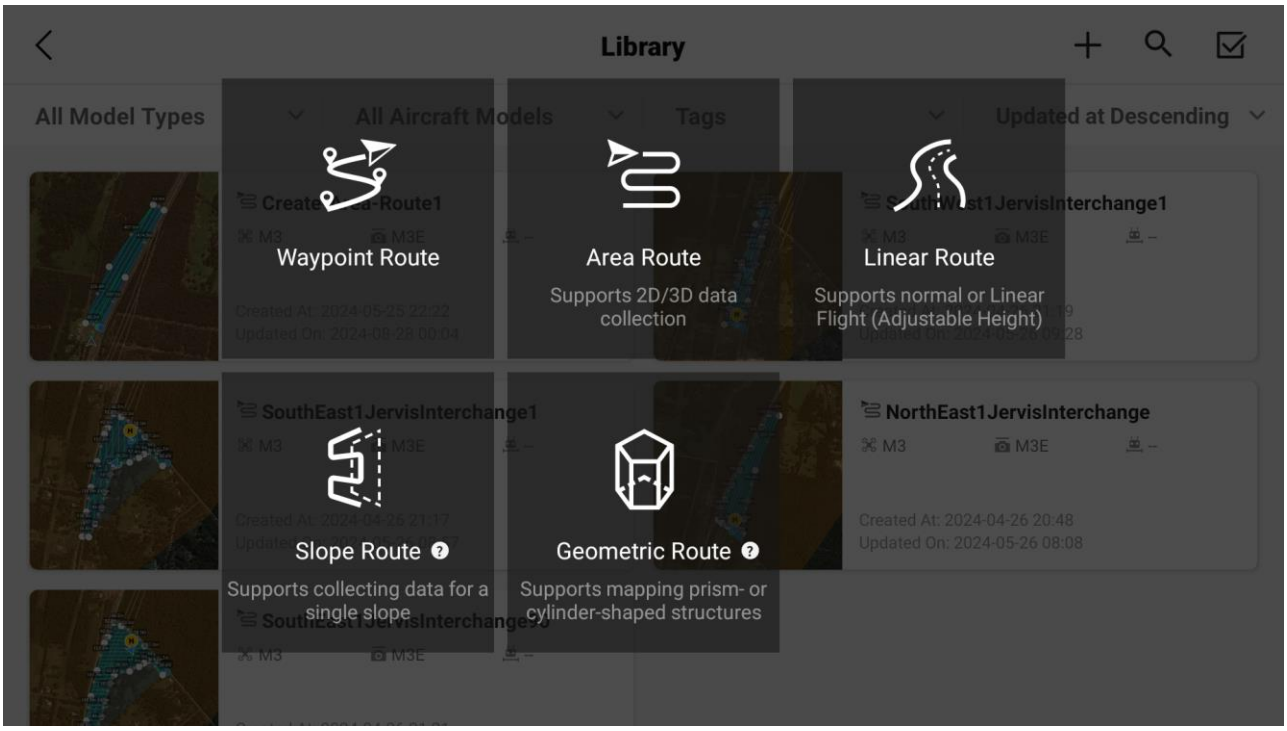
1. **Accessing Mission Planner**

- Open the DJI Pilot 2 app on your controller.
- Select "Flight Route" to view your library of previous plans. These plans can be sorted by capture type, aircraft model, tags, or recent updates.



2. Creating or Importing a Mission

- In the top-right corner, you'll see the plus button or "New Mission," "Search," and "Select" options.
- **New Mission**: Choose "Create a Route" to set up a new mission or "Import Route" to upload a pre-made plan.



- **Importing a Route**: Ensure the file is in DJI-compatible KMZ/KML format. Place it on a FAT32-formatted micro SD card, insert it into the controller, and select "Import Route" to access the file location.

Mission Creation and Planning

1. Starting a Mission

- After importing or creating a mission, tap it to open the mission planner, if unsure you will typically utilise the area route option.
- In the 2D map interface, tap points on the map to define the boundary around the target area.
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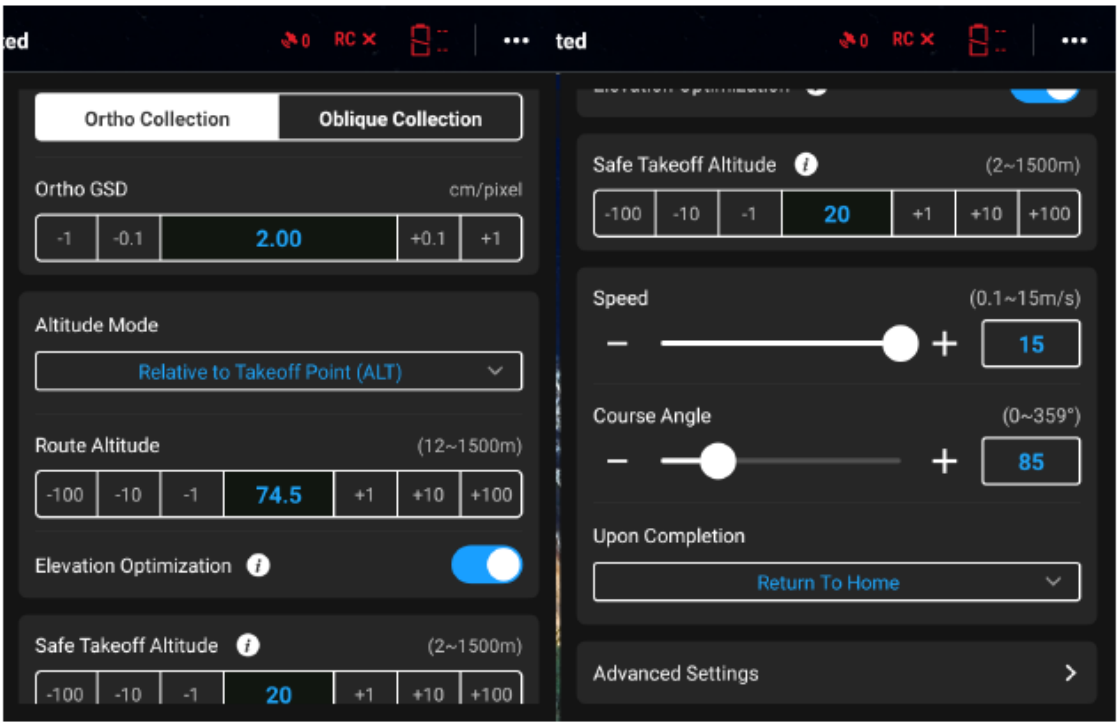
- Tap the blue checkmark to save the boundary, then select the UAV and camera model. Correct model selection is essential for setting accurate mission parameters.

2. Choosing Collection Mode

- **Orthomosaic Collection**: The UAV captures images in a straight-down (nadir) orientation, ideal for 2D mapping.
- **Oblique Collection**: Images are captured at a 45-degree angle, suited for 3D modeling of complex structures. This setting can be adjusted later if necessary.

Orthomosaic Collection Settings

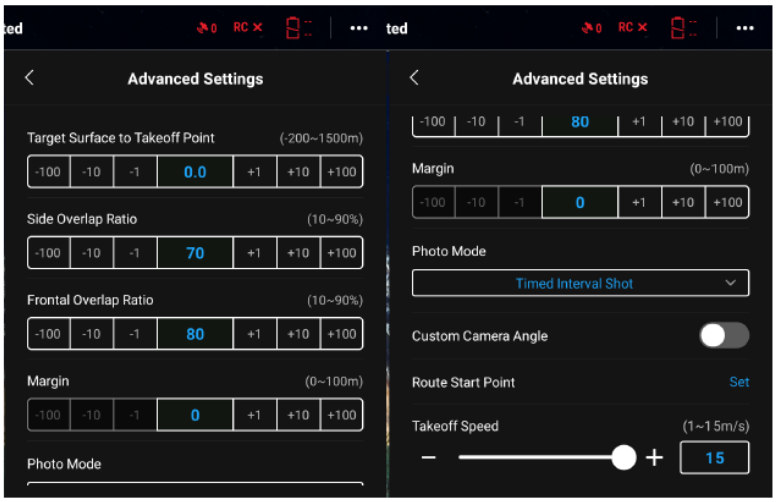
- **Orthomosaic GSD**: Adjust the Ground Sampling Distance (GSD) based on requirements specifications or altitude restrictions. Changing the GSD will also affect altitude and flight line spacing. Be mindful to ensure that the altitude you select allows safe operations within the area
- **Altitude Mode**: Options include "Relative to Takeoff Point (ALT)," "Above Sea Level (ASL)," or "Above Ground Level (AGL)." ALT is the default setting.



- **Real-Time Terrain Following** (AGL mode only): Available on Mavic 3, this setting uses onboard sensors or DSM files to adjust altitude in real-time, though it may not account for large structures or steep terrain changes.
- **Elevation Optimization**: Adds a 45-degree gimbal pitch photo capture at the center of the mission area to improve vertical accuracy.
- **Safe Takeoff Altitude**: Sets the initial height for the UAV before starting the mission path.
- **Speed**: Sets the maximum flight speed, which automatically adjusts based on the camera's interval limit.

Advanced Orthomosaic Collection Settings

1. **Target Surface to Takeoff Point**: Specifies altitude differences between takeoff and target areas, recommended if there's a significant elevation change when ALT mode is used.
2. **Side Overlap Ratio**: Defines the side image overlap between each flight line.
3. **Frontal Overlap Ratio**: Sets the overlap percentage for consecutive photos, essential for seamless stitching.

4. 

5. **Margin**: Adds a buffer around the mapped area to ensure coverage at the boundary edges.
6. **Photo Mode**: Choose between "Timed Interval Shot" (photos taken at time intervals) and "Distance Interval Shot" (photos taken at distance intervals). Distance intervals can maintain more consistent overlap at corners.
7. **Custom Camera Angle**: Adjusts the camera angle from nadir for unique perspectives if needed.
8. **Aircraft Rotation**: Sets the UAV's heading, either following the route or at a custom angle.
9. **Gimbal Pitch Rotation**: Sets camera pitch from -90° (nadir) to -30°, adjustable mid-flight.
10. **Route Start Point**: Designates the starting point of the mission.
11. **Take off Speed**: Sets speed to the first waypoint, typically set to maximum.

Oblique Collection Settings

Oblique collection adds additional captures at a 45-degree angle from each cardinal direction. This setup is ideal for full 3D coverage of complex projects.

1. Oblique Parameters

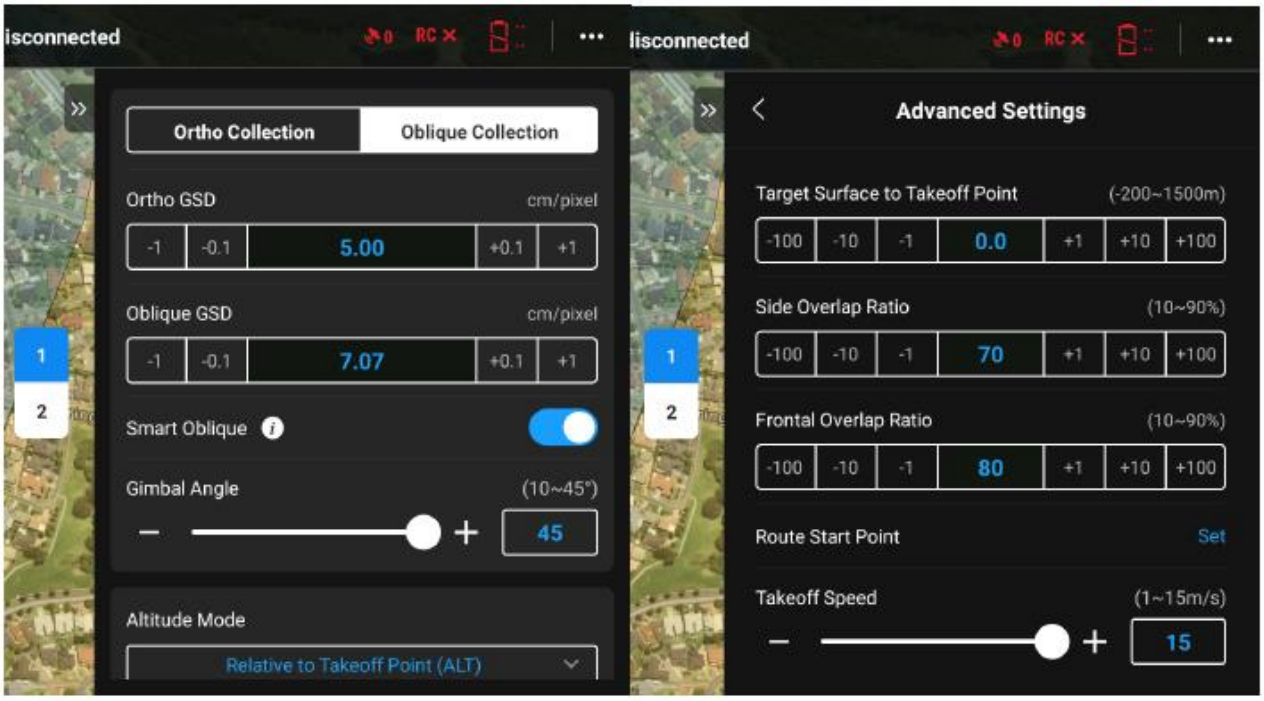
- **Orthomosaic GSD**: Sets Ground Sampling Distance for nadir images, based on altitude limitations or project needs.
- **Oblique GSD**: Defines GSD for angled captures, generally lower resolution due to the increased subject distance.
- **Smart Oblique**: Combines oblique and nadir tasks into a single task, reducing total flight time.
- **Gimbal Pitch Oblique**: Sets the oblique camera angle, defaulting to 45 degrees.
    - *(Use Figure 26: Oblique parameters including GSD, smart oblique, and gimbal pitch)*

2. Altitude Settings

- **Altitude Mode**: Options are the same as Orthomosaic: "ALT," "ASL," or "AGL."
- **Real-Time Follow**: Available in AGL mode, uses sensors for real-time altitude adjustments.
- **Route Altitude**: Sets the altitude for the mission, and adjustments here will also affect the GSD.

Advanced Oblique Collection Settings

- **Target Surface to Takeoff Point**: Specifies altitude difference between takeoff and target areas if ALT mode is selected.
- **Side and Frontal Overlap Ratios**: Sets side and front overlaps, critical for data consistency in stitching.
- **Route Start Point**: Defines where the UAV will begin the mission.
- **Takeoff Speed**: Sets the UAV's speed to reach the first point.



Saving and Starting the Mission

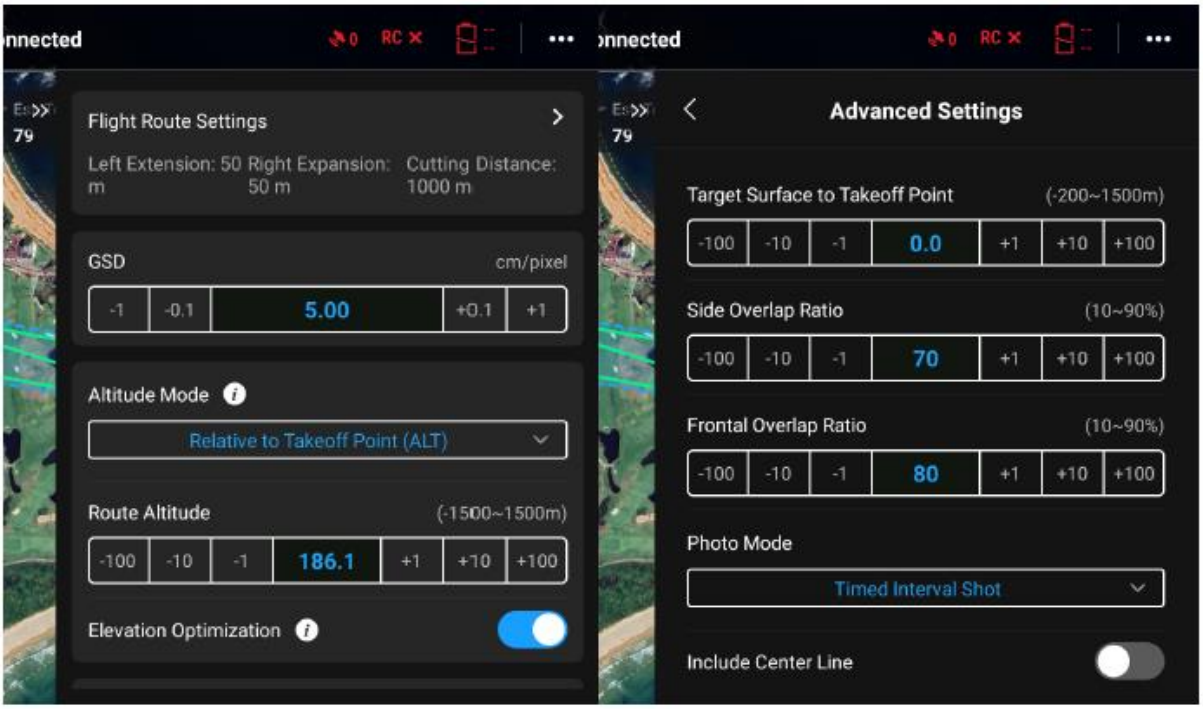
- **Save**: Tap the save icon in the top left to save your mission configuration.
- **Start Mission**: Press the blue play button to begin the Preflight and Mapping Checklist. Make sure "Dewarping" is enabled unless specified otherwise.
- The mission will then commence and capture all images, you will not need to take any pictures or manually fly, the RPAS will do this itself.

Additional Mission Types

1. Linear Route

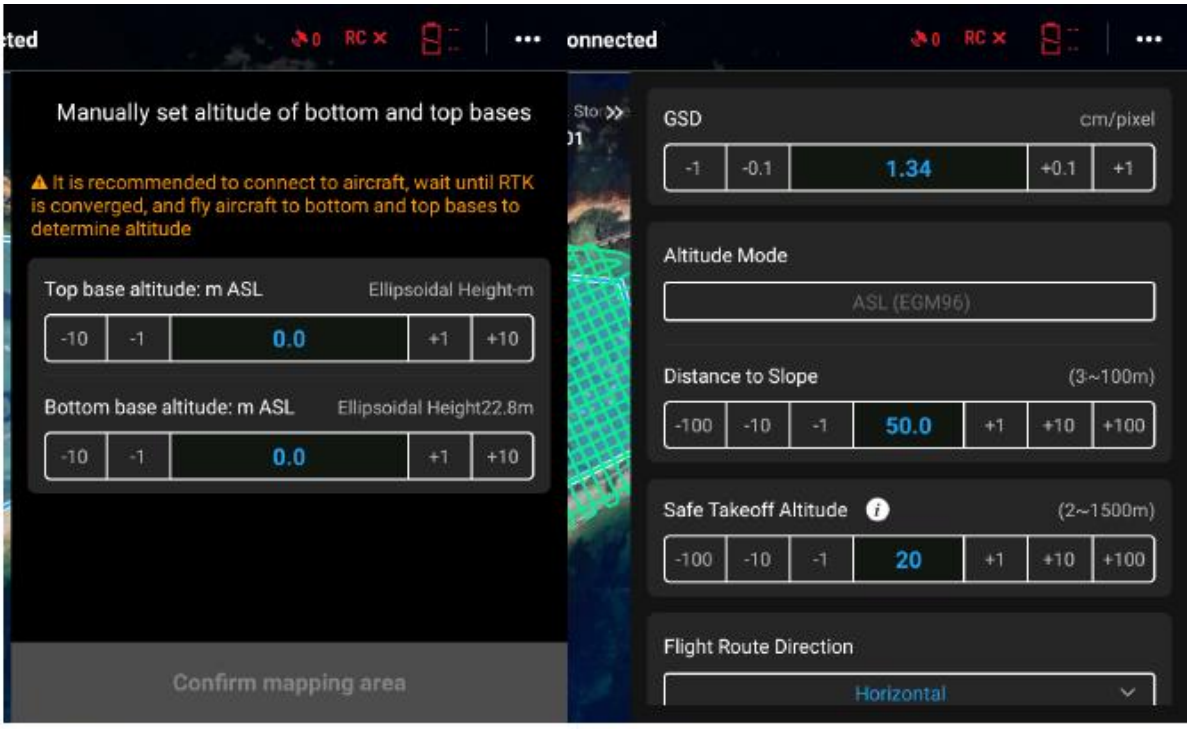
- Draw a line on the map to represent the route's center. Choose between "Zigzag" (multi-pass) or "Single Route" (single-pass) for coverage needs.
- **Extension Length**: Sets the buffer width on either side of the centerline.
- **Boundary Optimization**: Adds flight lines along the boundary to prevent gaps.
- **Flight Route Direction**: Choose between parallel or perpendicular to the centerline.

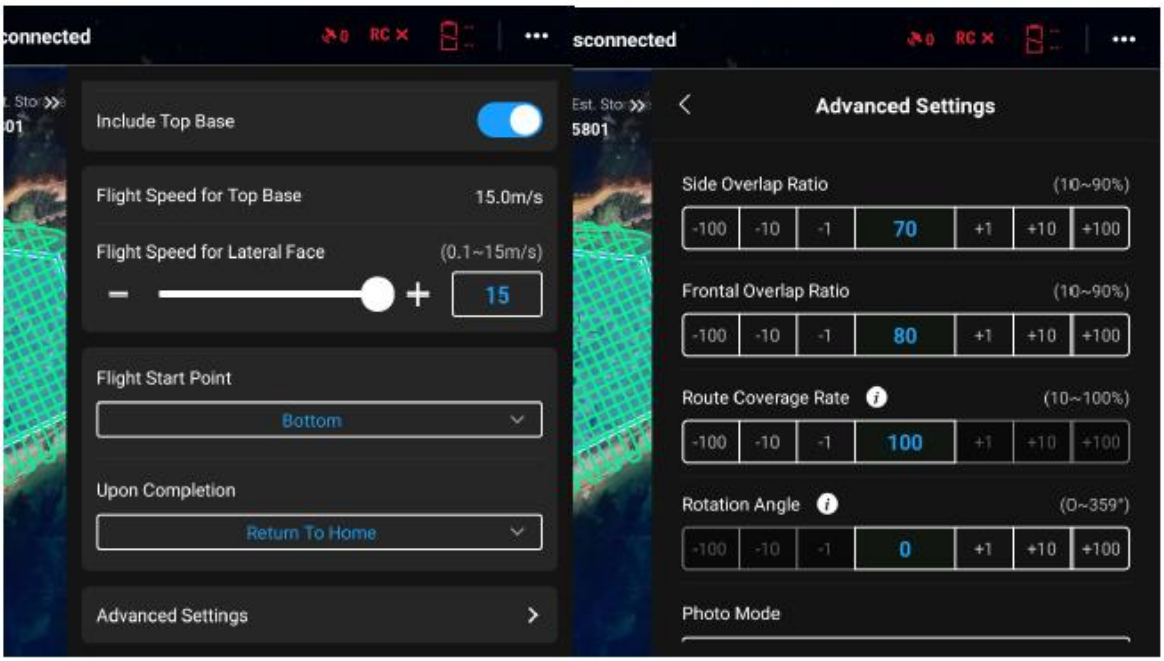




. Geometric Route

- Draw a polygon to define the area and set top/bottom altitude limits.
- **Distance to Slope**: Controls horizontal distance to vertical faces, depending on the GSD.
- **Flight Route Direction**: Choose horizontal or vertical for sloped surfaces.
- **Route Coverage Rate**: Adjusts coverage density for vertical surfaces.





21 Appendix 7: Searches

21.1 Overview

The use of RPA for assisting in searching for a missing person or object may be provided at the request of NSW Police.

21.2 Mission Profile

The use of RPA to assist with search missions. It is important that NSW Police initiate the request. Note that there are limitations in the successful use of RPA in search operations.

21.3 Task Profile

21.3.1 Mission planning

- Liaise with NSW Police and identify search area
- Plot area in RC using map view annotations
- Gather intel and consider human behaviour
- Determine search outputs (videos, photos, search data etc)

21.3.2 Determine camera settings

- Gimbal angle options (gimbal down for unconscious victim; gimbal 30* oblique for conscious victims). Be aware when using oblique camera may create a hiding spot/shadow behind an object as you pass by
- Camera selection options (recommended 2.0x zoom, 2.0x IR (SBS); optional 1.0x wide for rapid search or where IR will not be an option)

21.3.3 Height

- Must always fly above obstacle height
- Recommended 40-60m AGL, unless other factors are present.
- >60m for faster searches, but beware of reduced target size on the screen
- <40m for more detailed searches, but beware that the search will be slower

21.3.4 Speed settings

- Slow 2-3m/s – for heavily obstructed environments (e.g. forests)
- Fast 8m/s – for open environments or for a rapid search

21.3.5 Plan the route / search pattern

- If the last known position (LNP) of the victim is known, consider starting from that point and moving towards the outer edges of the search area in a uniformed pattern
- In other circumstances, start at furthest point from home
- Calculate correct line spacing distances:
    - When using 2.0x IR, 2.0x zoom (SBS) camera, use the 1:2 AGL spacing ratio (i.e. 50m AGL – 25m spacing between passes)

- If only using 1.0x wide camera, use 1:1 AGL ration during the mission (i.e. 60m AGL – 60m space between passes)
- Ensure you allow for a 'run-in' of gimbal angle
- Ensure search patterns maintains overlap during passes. Consider how any large changes in elevation may change height AGL and therefore create overlap gaps

21.3.6 Locating person / object of interest

- Pause an automated flight plan
- Confirm item of interest is the target
- Drop pin/record GPS (ideally from above)
- Capture screen shot (including GPS
- Advise NSW Police

21.3.7 Data

- Liaise with NSW Police when planning/performing search missions to ensure data type
- Save data and provide to NSW Police in the format required

22 Appendix 8: Structure Fires

The use of RPA for capturing imagery at structure fires.

22.1 Mission Profile

The use of RPA to assist with monitoring fire progress, identification of exposures and identifying hot spots.

22.2 Task Profile

22.2.1 Mission planning

- Locate yourself close to the IC point if possible
- Maintain VLOS – do not fly through smoke
- Be aware of flying over populous areas
- Consider use of 360 panorama images

22.2.2 Determine camera settings

- Unless specific photos / videos are needed (e.g. ignition point) guidelines for fireground photos:
    - Shoot from 45* elevation (up and out)
    - Shoot from outside corners of the structure
    - Include property location information such as a house number or letterbox if possible
    - Include exposure risks, water supply and appliances if possible (for debrief)

- Keep videos to around 20 seconds
- Use “tripod” mode to record better video

22.2.3 Data

- Liaise with IC when planning/performing missions to determine required data type
- Save data and provide to IC in the format required

23 Appendix 9: Bush & Grass Fires

23.1 Overview

The use of RPA for capturing imagery at bush and/or grass fires.

23.2 Mission Profile

The use of RPA to assist with monitoring fire progress, identification of exposures and identifying hot spots.

23.3 Task Profile

23.3.1 Mission planning

- Be aware of other aircraft operating in the area. Always monitor radio and maintain separation. If in doubt, land the aircraft
  - Locate yourself close to the IC point if possible, however be aware of:
  - Actual and predicted fire progression
  - Impact of smoke
- Access / egress and escape routes for firefighting appliances
- Maintain VLOS – do not fly through smoke

Be aware of flying over populous areas

23.3.2 Determine camera settings

- Unless specific photos / videos are needed (e.g. ignition point) guidelines for fireground photos:
  - Keep videos to around 20 seconds
  - Use “tripod” mode to record better video

23.3.3 Data

- Liaise with IC when planning/performing missions to determine required data type
- Save data and provide to IC in the format required

24 Appendix 10: UAV Operations Around the Fire Scene

Overview

This appendix provides detailed guidelines for UAV pilots operating in close proximity to fire scenes. It outlines essential considerations to ensure UAV safety and effectiveness while supporting the NSW Rural Fire Service (RFS) in managing fire incidents.

Detailed Guidelines for UAV Operations

1. **Awareness of Current Fire Conditions and Predicted Winds**
    - UAV pilots should stay continuously updated on current fire status and potential changes in fire behaviour, including hotspots and spreading patterns, by working closely with the Incident Controller (IC) for real-time updates.
    - Monitor predicted wind patterns carefully, as shifts in wind direction can dramatically influence fire behaviour, intensity, and smoke spread. Pilots should be prepared to adjust flight paths or abort missions if wind conditions pose risks to UAV stability or personnel safety.
    - Consider wind speed variations at different altitudes, which may impact UAV control and data quality.
2. **Location of Primary and Secondary Landing Zones**
    - Identify and establish primary and secondary landing zones that are away from both current and projected fire edges. Landing zones should remain clear of smoke plumes and other environmental hazards to ensure safe and accessible UAV recovery.
    - Position landing zones with ample clearance from active fire perimeters and potential ember attack zones. Pilots should continuously monitor fire movement and be prepared to relocate landing zones if conditions change.
    - Secondary landing zones should serve as contingencies in case the primary zone becomes unsafe due to fire progression, smoke, or debris.
3. **Awareness of Air Temperatures and Radiant Heat from Fires**
    - High air temperatures and intense radiant heat near fires can impair UAV performance and sensor functionality, impacting battery life, internal component temperatures, and increasing the risk of mid-air failure.
    - UAV pilots should maintain a safe distance from high-temperature areas and continuously monitor UAV temperature readings, particularly during prolonged flights. Radiant heat levels should be factored into flight planning to avoid overexposure to extreme temperatures.
    - Recognise that UAV operating limits may be reached sooner in hot conditions, and be prepared to adjust flight durations to protect equipment and data integrity.
4. **Awareness of Airborne Particulates and Smoke Ingestion**

- Airborne particulates, including embers, ash, and debris, pose serious risks to UAV sensors and sensitive components. Particulates may obstruct camera lenses, impact sensor accuracy, and abrade propellers and rotors, reducing UAV effectiveness and lifespan.
- Smoke ingestion can damage the UAV's internal components by introducing contaminants into cooling systems and sensors, potentially leading to premature wear or failure. Avoid flying directly downwind of active fires or within smoke plumes to minimise exposure.
- Ensure thorough inspections of UAVs for particulate buildup after each mission. Maintenance should focus on sensors, filters, and air intake areas to prevent long-term damage.

5. **Mitigating Risks to UAV Sensors and Aircraft Components**
   - Use protective measures, such as lens filters and covers where possible, to guard against particulate accumulation during flights.
   - If available, consider using specialised UAVs or accessories designed for fire and smoke environments to improve resilience in harsh conditions.
   - Schedule regular assessments and replacement of vulnerable components (e.g., filters, propellers) exposed to particulate-heavy environments to ensure reliable UAV operation throughout fire scene activities.